

Who is responsible for security in an IaaS cloud service model?

- A) The customer only**
- B) The cloud service provider only**
- C) The ISP used by the customer**
- D) Both the customer and cloud service provider**
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Explanation

In an Infrastructure as a Service (IaaS) cloud service model, both the customer and cloud service provider (CSP) share the responsibility of security for the cloud deployment. Neither the customer nor the CSP take complete responsibility for the security of the cloud deployment. In a shared responsibility model, the customer is responsible for security in the cloud and the CSP is responsible for security of the cloud. The CSP's responsibility includes the security of the data center, physical servers, and services. The customer's responsibility includes the security of the workloads. To summarize, the following are the responsibility of the CSP:

- Data center security
- Virtualization infrastructure
- Software and hardware infrastructure
- Networking infrastructure
- Global infrastructure, including Availability Zones, Regions, and Edge locations

Customers are responsible for securing all the data that they create and store in the cloud. The following are the responsibility of the customer:

- Patching and configuration of operating systems running on virtual machine instances
- Setup of security groups
- Administration of user accounts and permissions

The Internet service provider (ISP) that provides Internet services to the customer is not responsible for security of a cloud deployment. The ISP only provides the necessary Internet connectivity for access to cloud resources but is not involved with securing either the underlying cloud infrastructure or of applications and data stored on a cloud. Additionally, when connecting to a cloud account over the Internet, security mechanisms like virtual private networks (VPNs) may need to be used for securing data transfer.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is IaaS?](#)

An organization is building a modern, unified analytics platform using Google Cloud. It must decide between using a data lake or a data warehouse for its data analytics needs. Which of these would be an appropriate business use case for using a data warehouse?

- A) Known datasets need to be analyzed and preset questions answered**
- B) There is no clarity on the kinds of analysis that may need to be done on the data**
- C) There is a need to discover data across various data types**
- D) There is a need to explore insights using existing resources**
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Explanation

When known datasets need to be analyzed and preset questions answered then a data warehouse would be the appropriate choice for an organization. A data warehouse is a massive repository that contains a company's data gathered from various different sources. This data includes current data as well as data that is historical. A data warehouse is used for business intelligence (BI) activities like complex analytics which can lead to making informed management decisions. The sources of data for a data warehouse include customer relationship management (CRM) systems, automation systems for marketing, applications for transactions, and log files from applications. A data warehouse can be used for performing analysis in an ad hoc manner or for creating custom reports.

The other options are all characteristics of business use cases that can benefit from using a data lake:

- There is a need to discover data across various data types
- There is a need to explore insights using existing resources
- There is no clarity on the kinds of analysis that may need to be done on the data

A data lake allows you to store both structured, semi-structured, and unstructured data from multiple sources in a single place. This data can come from a variety of sources, including multimedia, social media, hard disks, servers, Internet of Things (IoT) devices, clickstream, databases, and transactional information. A data lake provides security and scalability and enables companies to perform ingestion of data from various sources including edge computing, cloud, or on-premises systems. The data can be stored in its original format and processed as a batch or in real-time. This data can then be analyzed using tools like Python, SQL, and R.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Services.google.com > Build a modern, unified analytics data platform > Data warehouse or data lake?](#)

Your company needs to assess its readiness for adopting the Google Cloud platform. For this, you need to assess its cloud readiness. Which theme of the Google Cloud Adoption Framework covers the ability of a company to leverage services that are native to the cloud to reduce overhead in its IT operations?

- A) Lead**
- B) Learn**
- C) Scale**
- D) Secure**
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Explanation

The correct option is scale. This is the theme that covers the level to which a company uses cloud services, provisions workloads, performs automation, and maintains the applications it needs for its business. This theme covers how a business uses native cloud technologies, like serverless computing and automation, to improve its operations and scalability.

The other options are incorrect as they do not measure a company's depth of involvement in leveraging cloud-native systems to improve its operations. The other options are also a part of the themes of the Google Cloud Adoption Framework (CAF) and address different aspects of the cloud readiness of a company. The CAF is a way for an organization to assess its cloud readiness and find actions to prepare itself for successful cloud adoption. The CAF provides a structure of people, processes, and technology that are involved in a company planning a cloud adoption. There are three components to Google's CAF:

- Three maturity phases
- Four cloud adoption themes
- Seventeen epics

A component of the CAF is the four themes:

- Learn – This measures the learning systems in place for an organization to upskill its IT workforce. It also includes the ability of a company to leverage partners who are skilled in cloud technology to augment its existing workforce.
- Lead – This measures the level of support that the IT workforce receives from the company's leadership for cloud migration. It also covers the level of motivation, collaboration, and cross-functionality of the teams. The team structures, executive sponsorship, and budgeting for cloud projects are also covered in this theme.
- Scale – This measures the depth of usage of services that are native to the cloud for reducing overhead in operations, as well as the level of automation you perform for policies and processes which were operated manually. This theme covers how cloud services and resources are provisioned, capacities of workloads distributed, and updates for applications performed.
- Secure – This measures the level of protection in place for your environment from malicious access using a security system that has multiple layers and is identity-centric. This covers the access control systems in place as well as the security strategies and technologies that you use to protect your services.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Question #4 of 60

Question ID: 1497135

Your company is in the process of adopting Google AI technology for its business growth. You are evaluating fairness and security in Google's AI and machine learning differentiators. Which of these is a recommended best practice for Google's stipulated responsible AI?

- A) Keep all the features in your ML model, including redundant ones**
- B) Train models to create automatic inferences based on detected correlations**
- C) Perform engagement with multiple users in the development process
- D) Conduct short-term fixes to all issues for swift problem solving**

Explanation

Performing engagement with multiple users in the development process is a recommended best practice for Google's stipulated responsible artificial intelligence (AI) framework. This is part of a best practice of utilizing a design approach that is centered on actual users. The effectiveness of an AI system's predictions and decisions can be gauged from the experiences of actual users. Incorporating inputs from a wide range of users and use cases helps enrich the project with a range of perspectives from users. This eventually benefits the users that actually utilize the technology. The system needs to have enough built-in disclosures to users so that a satisfying user experience can be achieved.

Keeping all the features in your machine learning (ML) model, including redundant ones, is not a recommended best practice for Google's stipulated responsible AI. The recommended practice is to use the ML model, which is simple and meets goals for performance without using redundant features. The raw data used by an ML model needs to be analyzed, and if the model uses sensitive data, then that data should be understood within privacy mandates for the data. The data needs to be checked for errors, bias, and training-serving skews to ensure the optimal performance of the model.

Training models to create automatic inferences based on detected correlations is not a recommended best practice for Google's stipulated responsible AI. You need to ensure that the ML model does not make casual inferences based on correlations that it detects. You also need to be aware of the limitations of the model based on its training data's scope and range. Communicating these limitations to users can help with getting useful feedback about the AI system from users.

Conducting short-term fixes to all issues for swift problem solving is not a recommended best practice for Google's stipulated responsible AI. Short-term fixes may save time but do not offer solutions that are beneficial in the long-term. For this reason, it is important to create a balance between using short-term fixes and long-term solutions. Also, you need to compare the candidate ML model with the deployed version to identify the ways they differ, and to understand how the update will alter the user experience and system performance of the AI application.

Some additional best practices include using metrics like user feedback, false negatives and positives, product health reports, and system performance data for assessing monitoring and training of the ML model. Also, testing the system with both integration tests as well as unit tests to ensure that the performance of the AI system is as expected. This involves studying how the individual components of the ML model work with other components of the application. Also, continuous monitoring of the model needs to be done to ensure the model incorporates feedback from the real-world and user experiences.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Ai.google/responsibilities > Responsible AI practices](#)

Question #5 of 60

Question ID: 1495288

Your operations team spends a lot of time on repetitive tasks. You want to bring greater efficiency into your Google Cloud operations by reducing toil. What would be the best strategy to achieve this goal?

- A) Use more observability tools**
- B) Perform encryption of data at rest**
- C) Create new SLAs with users
- D) Halt deployments**

Explanation

You should use more observability tools to reduce toil. Toil comprises manual and repetitive tasks, which should be automated so that team members can focus their time on higher value work. By using observability tools like Cloud Tracing, Cloud Monitoring, and Cloud Logging, you can gain more insight into services and ensure their correct automation. Creating more automation of services can help reduce toil in an organization, which boosts work efficiency and productivity. A Site Reliability Engineering (SRE) recommended best practice is for operations teams to look into issues with services and attempt to identify the reasons for failure. For this, the teams should gain more knowledge on how a service operates.

You would not perform encryption of data at rest to reduce toil. Encryption at rest of stored data is a way of implementing security for data stored on the cloud infrastructure. Google utilizes multiple layers of encryption for the protection of data at rest. Google's storage infrastructure encrypts all user data by default before it is stored on physical storage. This way, encryption is performed by Google Cloud at the storage or application layer of the cloud infrastructure.

You would not create new service level agreements (SLAs) with users to reduce toil. SLAs are an agreement between the service provider and customer that specifies the level of availability that the service will have. Reduction of toil involves creating more automation of systems, which removes the need for manual involvement of team members.

You would not halt deployments to reduce toil. Halting deployment is not done for toil reduction but is done in situations where an error budget is consumed. When new features are added to a service, new bugs may be introduced, which can eat into an existing error budget. Once an organization reaches a stage where the error budget for a given period is exhausted, then management will need to halt the development of new features until the error budget is restored.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Question #6 of 60

Question ID: 1495899

You work for a software development company that creates application software for desktop and mobile users as well as custom designed operating systems. The company now needs to migrate to a cloud-based system that can provide all the physical hardware it needs. The company will still utilize its self-owned software and applications.

Which cloud service model would you use based on this requirement?

- A) SaaS**
- B) IaaS**
- C) DaaS**
- D) PaaS**

Explanation

You would deploy the Infrastructure as a Service (IaaS) model. This provides the necessary hardware via the cloud that the company can then utilize to run its applications and even install a customized operating system (OS). The IaaS model allows the use of barebones computing resources, such as physical hardware and networking, so that consumers can install the OS of their choice and run any applications they wish. The advantage of this is that a company's existing data center hardware resources can be completely replaced with a cloud-based solution. The IaaS model provides networking, storage, processing, and other basic computing resources for consumers.

The other options are incorrect because they do not allow the installation and use of a customized OS and they are better suited for needs related to consumers who want an existing software environment.

For the exam, you need to know some key concepts related to cloud computing. Cloud computing is the provision of computational services made available to consumers by cloud providers based on their need or consumption. These services cover both the use of platforms and applications as well as physical hardware resources. There are three primary cloud service models that provide services accessible by Web browsers, cellphones, and similar devices. The three primary cloud service models are PaaS, SaaS, and IaaS.

The Platform as a Service (PaaS) model allows consumers to install their own applications onto a cloud platform. The provider supplies the necessary OS and physical hardware. This allows consumers to quickly deploy their own applications without the burden of first acquiring the necessary physical resources and OS to make it happen. The PaaS model provides compute services, OS, networking, storage, and all the necessary hardware.

The Software as a Service (SaaS) model covers the use of software applications by consumers who are running on a provider's cloud infrastructure. These applications can be accessed using thin client interfaces, including cellphones and Web browsers. An example of this is email accessible from a Web browser, such as Gmail. The advantage of this model is that a consumer can utilize the application without the need to physically install the application or be concerned about the infrastructure needed to run it. The cloud provider takes full responsibility for running the application and managing it. Other examples of utilities that fall under the SaaS model include Enterprise Resource Planning (ERP), software development programs, Customer Relationship Management (CRM) systems, and Human Resources (HR) applications. Software applications for storage, networking, and processing also comprise the SaaS model.

Other cloud service models include:

- Communications as a Service (CaaS) – This service model includes video conferencing, chat, voice calling, emailing, and other means of collaboration.
- Anything as a Service (XaaS) – This service model includes the provision of complete IT solutions as a consolidated unified package covering end-to-end services.
- Desktop as a Service (DaaS) – This service model provides a virtual desktop that can be accessed by a variety of devices, including laptops, tablets, and cellphones. The desktop has all of the applications typically required by business workflows, including word processors and spreadsheets.
- Business Process as a Service (BPaaS) – This service model provides applications supporting routine business operations, such as shipping, orders, inventory, and payroll.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is IaaS?](#)

Question #7 of 60

Question ID: 1495166

Your company is looking at ways to change its operation and resource utilization so that it excels at delivering the maximum possible value to customers. For this, it is looking into performing digital transformation using cloud technology.

Which of these is a key part of the journey to digital transformation?

- A) Keeping team collaboration focused within each company department**
- B) Segregating customer feedback from product development activities**
- C) Guiding business decisions using insight from data analytics tools**
- D) Using legacy on-premises systems for quickly deploying new global applications**

Explanation

Guiding business decisions using insight from data analytics tools is a key part of the journey to digital transformation. Experimentation with new technologies and concepts can help a company gain more insight into market needs and customer preferences. This experimentation can then be analyzed using advanced data analytics tools to make the right business decisions for growth.

Segregating customer feedback from product development activities is not a key part of the journey to digital transformation. This is because customer feedback needs to be incorporated into product development efforts and not segregated from them so the company can create offerings that better serve customers. Focusing on customers and their preferences through analysis of customer usage data can help create products that provide more value to customers and lead to greater business growth.

Keeping team collaboration focused within each company department is not a key part of the journey to digital transformation. Digital transformation involves using tools to share knowledge and collaborate across a company's organizational boundaries. Team collaboration and knowledge sharing eventually leads to more autonomy in decision making and creates a more dynamic workforce.

Using legacy on-premises systems for quickly deploying new global applications is not a key part of the journey to digital transformation. Digital transformation involves using modern technology that has the flexibility to enable you to deploy applications onto the cloud swiftly and manage them efficiently.

Digital transformation involves utilizing new technology to change the design and definition of a company's relationships with its partners, customers, and employees. Digital transformation involves using new digital systems including private, public, and hybrid clouds to change existing culture, work processes, and customer experiences with an aim to meet the changing dynamics in businesses and markets. Some key benefits of digital transformation include:

- Modernized infrastructure – This includes using serverless systems, containers, and global cloud networks for an efficient and effective IT infrastructure.
- Better data management – This include using new systems and tools to manage massive amounts of data which originate from various devices and sources.
- Stronger insights – This covers using advanced business analytics tools to provide useful data insights.
- Faster collaboration – This involves dismantling the traditional team silo concept where a team is cut off from other teams and instead using new tools that allow teams to collaborate across various global locations swiftly. Faster collaboration helps create value and produce results for customers more quickly.
- Finding business solutions – This involves using new systems, processes, and technologies for solving business problems and identifying ways to improve how businesses interact and work with their customers.
- Savings on costs – This involves using tools for making processes more efficient and identifying the high and low performance aspects of a business accurately.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is Digital Transformation > What are the benefits of digital transformation?](#)

Question #8 of 60

Question ID: 1497153

You are a consultant for a stock trading company that is setting up its Google Cloud environment. You need to guide the team on some key issues that the company needs to be aware of when managing cloud environments. Which of these concerns should you specify? (Choose two.)

- A) Less cloud usage transparency

✓
- B) More overhead using cloud services

✗
- C) Lack of cost predictability

✓
- D) Lack of on-premises servers

✗
- E) Security of servers housing virtual machines

✗

Explanation

You should specify lack of cost predictability and less cloud usage transparency as key issues the company should be aware of. Lack of cost predictability can be an issue for companies managing cloud environments. A customer application may experience surges in user traffic alternated with periods of low levels of traffic. Due to the unpredictable nature of application usage, it can become difficult for a cloud administrator to specify resource limits. Setting resource limits too low can lead to application performance issues during peak usage, and provisioning more resources than required can lead to wasted capital. Less visibility and transparency in cloud usage leads to the wastage of cloud expenditure due to unutilized resources. By using cloud monitoring tools, a cloud consumer can identify resources that are unused and deprovision them so that they do not contribute to the overall billing. This helps reduce unnecessary costs for a cloud deployment.

The lack of on-premises servers is not a concern when managing a cloud environment because the set-up of an on-premises data center and its operation are not related to a cloud environment. To increase the computing power of a cloud deployment, you can perform scaling either manually or automatically, which provisions more virtual servers when required.

More overhead using cloud services is not a concern when managing a cloud environment. By using fully managed services on Google Cloud, you can reduce your operating costs by removing all the overhead that would be required for various tasks, which include resource provisioning, backups, updates, and patching. This reduction in costs allows you to better utilize your capital to grow your business.

The security of servers housing virtual machines (VMs) is not a concern when managing a cloud environment. This is the responsibility of the cloud service provider (CSP). As per the shared responsibility model of cloud environments, the customer is responsible for securing the data and applications that are hosted on the cloud, and the CSP secures the entire cloud infrastructure, which includes networking systems and physical servers that house VMs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com >A guide to financial governance in the cloud](#)

Question #9 of 60

Question ID: 1495932

A company wants to find a way to find anomalies in shopping transactions of users that browse its site and mobile app through a secure API-based system that is fully managed and scalable. Which Google Cloud service should you use for this?

- A) Looker

✗
- B) Dataproc

✗
- C) Firestore

✗
- D) Payment Gateway

✓

Explanation

You should use Payment Gateway for this scenario. With Payment Gateway, you can perform processing of payments that are application programming interface (API)-based. It can scale up to cater to unpredictable surges in usage as well as be extended to third-party insights and anomaly finding tools through APIs. It also allows for payment processing across multiple regions of Google Clouds global infrastructure. Payment Gateway also has built-in tools that allow you to monitor, debug, and trace API-based purchases with high availability and less overhead. You can also export data of transactions into a project on Google Cloud using an export API.

You would not use Firestore for this scenario. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Looker for this scenario. Looker is a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications to provide embedded analytics. Looker can be used to create alerts to proactively send timely insights related to various business data. It also has dashboards that are interactive, which allow you to analyze data in real-time through a trusted data model.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Payment Gateway](#)

Question #10 of 60

Question ID: 1495947

Your company has a photo sharing service that is heavily in use. You have created an SLO for the service. You need to find a way to check the status of your SLO and also have the error budget computed automatically. Which Google Cloud service would be suitable for this requirement?

- A) Looker**
- B) Cloud Monitoring**
- C) BigQuery**
- D) Cloud Vision**
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Explanation

Cloud Monitoring can be used to create and use service level objectives (SLOs), service level indicators (SLIs), and error budgets. You can specify an SLO for a month, and Cloud Monitoring computes the corresponding error budget automatically. It can be used to monitor the complete infrastructure of a cloud deployment. It provides a view into the uptime, health, and performance of cloud systems. For this, Cloud Monitoring gathers relevant metadata, events, and metrics from sources that include uptime probes, services, app instrumentation, alerts, notifications, and other components of applications. It allows you to stop issues from happening, and it allows you to identify trends. It also reduces overhead related to monitoring through a service that is integrated and provides metrics, dashboards, and alerts all in one place. Cloud Monitoring provides SLO monitoring and metrics collection for virtual machines (VMs) (using Ops Agent) and Kubernetes (using Managed Service for Prometheus).

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence, and geospatial analytics ready for use, and has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This allows you to perform queries on petabytes of data within minutes.

You would not use Looker for this scenario. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications to provide embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an application programming interface (API). It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Monitoring](#)

Question #11 of 60

Question ID: 1497134

You are designing a machine learning (ML) production system for your company. You need to create the ML pipeline in a way that ensures that models meet performance expectations. How would you do this? (Choose two.)

- A) Perform manual updates to ML models post-deployment**
- B) Maintain a tactical maturity phase for ML applications**
- C) Keep the model validation process at the end of the pipeline**
- D) Keep a one-time training for the ML model pre-deployment**
- E) Keep the data validation process at the beginning of the pipeline**
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Explanation

Data validation measures need to be placed at the beginning of the pipeline and model validation needs to be placed at the end. Data validation is used to verify the data types and schemas for the fresh data, which is used for retraining an existing machine learning (ML) model. Model validation checks the performance of an ML model's predictions and ensures that the performance is better than its predecessor.

Keeping a one-time training for the ML model pre-deployment will not help meet performance expectations for the ML model. This is because ML models need continuous training (CT). CT helps validate all the components and code of ML systems and checks existing models and schemas. The purpose of CT is to help ML models adapt to changes in data before the models are deployed. This is done automatically. To automate CT reliably, you need to place validation and quality checks for ML training into the existing ML pipeline.

You would not maintain a tactical maturity phase for ML applications as this would not meet the enterprise-level needs of a ML production system. The tactical maturity phase is when a company is only exploring the possibilities of using artificial intelligence (AI) and ML technology. This maturity level is characterized by the use of limited computing resources like data science virtual machines (VMs) for local data processing and training ML models. The tactical maturity level also involves the exploration of AI possibilities through use cases for proof of value (POV) demonstrations. This maturity level sees controlled costs for ML exploration with hardware and compute resources that are dedicated for use by selected data scientists. The exploratory analysis of data is done using ready-for-use AI systems and tools for data wrangling.

You would not perform manual updates to ML models post-deployment. This is not a suitable approach because manual updating can lead to errors and is not a feasible solution in an enterprise that may have a massive number of deployed ML models that need regular updates of their data as well as the implementation of new ML concepts. An automated pipeline-based approach would be ideal for updating ML models. This approach is used in transformational maturity, which involves orchestration and automation of ML and data pipelines end-to-end.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Services.google.com > Google Cloud's AI Adoption Framework](#)

Question #12 of 60

Question ID: 1495191

Your company has various multi-cloud and hybrid cloud environments that store its data. You need to find a way to populate this data lake with data from all of these sources. Which service should you use to perform data ingestion for the data lake?

- A) Cloud Trace

✖
- B) Cloud Data Fusion

✔
- C) Dataproc

✖
- D) Cloud Storage

✖

Explanation

You should use Cloud Data Fusion for the process of data ingestion into the data lake. Cloud Data Fusion is a fully managed service that provides scalable integration for cloud-native data. It allows users to deploy Extract, Transform, Load/Extract, Load, Transform (ETL/ELT) data without requiring coding by using a visual interface. Cloud Data Fusion is created using the Centralized Data Analysis Platform (CDAP) data analytics platform, which is open source. This allows for the portability of data pipelines and integration for multi-cloud and hybrid cloud deployments that enable users to remove existing solos and access previously inaccessible data. Additionally, Cloud Data Fusion provides several preset transformations that can be used for stream and batch processing tasks. This boosts collaboration and productivity by allowing connections and transformations to be shared among users.

You would not use Cloud Storage for data ingestion. This is because it is an object storage service and not a data ingestion service. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

You would not use Dataproc for data ingestion. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, ETL operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Cloud Trace for data ingestion. Cloud Trace is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Cloud Data Fusion](#)

Question #13 of 60

Question ID: 1495275

Your company is using Google Cloud and needs to find a way to incorporate its existing employee accounts in Active Directory with Google Cloud. What will you need to do to accomplish this?

- A) Maintain Google Cloud identities for each employee manually

✖
- B) Configure a lifecycle policy for data objects on the cloud

✖
- C) Federate user identities between Cloud Identity and Active Directory

✔
- D) Make a new Active Directory account for each employee

✖

Explanation

You will need to federate user identities between Cloud Identity and Active Directory. Google Cloud Identity is used for managing access and authentication. You can use Cloud Identity to manage Google identities in the company using automation. Cloud Identity allows you to link the lifecycles of Google identities to the identity management systems that are already functioning in an organization. The federation of user identities between Active Directory ensures that Google identities are maintained in an automated manner. By provisioning a user in Cloud Identity and Active Directory, you can enable single sign-on (SSO) for authenticating the user. SSO is a system in which a user can utilize one set of credentials and log into multiple independent systems.

You would not maintain Google Cloud identities for each employee manually. This is not an optimal way of managing identities as employees already have an account with an identity store like Active Directory. Manually maintaining Google identities would add large overhead for management. Instead, opting for identity federation is a more appropriate approach in this scenario.

You would not configure a lifecycle policy for data objects on the cloud. This does not help manage access for users on Google Cloud, which is what the scenario requires. Configuring a lifecycle for data objects is a way of setting up storage in a way that suits your company's data storage needs. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

You would not make a new Active Directory account for each employee. This is because the existing employee Active Directory accounts can be successfully incorporated into Google Identities using identity federation. Google also provides the Google Cloud Directory Sync tool that synchronizes key groups and users regularly from Active Directory to Cloud Identity.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Services.google.com > Google Cloud security foundations guide > Authentication and authorization](#)

Question #14 of 60

Question ID: 1495195

A company has an on-premises PostgreSQL server database system and is looking for a cloud-based alternative that is fully managed so the company can remove overhead costs associated with replication, maintenance, and updates. Which of these solutions should you recommend to meet this requirement?

- A) Cloud Data Fusion

✖
- B) Cloud SQL

✔
- C) Cloud Trace

✖
- D) Dataproc

✖

Explanation

You should recommend Cloud SQL for this scenario. Cloud SQL provides a relational database system that is fully managed and works with SQL Server, PostgreSQL, and MySQL. It provides features like flags for configurations, rich extension collections, and ecosystems for developers. Cloud SQL brings down the maintenance costs of databases and provides reliable operations supported by a site reliability engineering (SRE) team that is available around the clock. It has automation built-in for tasks like managing storage capacity and provisioning for databases. It can also be integrated with Google Kubernetes Engine (GKE) and BigQuery while also providing observability of your data using Cloud SQL Insights.

You would not recommend Cloud Trace in this scenario because it is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

You would not recommend Cloud Data Fusion in this scenario because it is a fully managed service that provides scalable integration for cloud-native data. It allows users to deploy Extract, Transform, and Load/Extract, Load, and Transform (ETL/ELT) data without requiring coding by using a visual interface. Cloud Data Fusion is created using the Centralized Data Analysis Platform (CDAP) data analytics platform which is open source.

You would not recommend Dataproc in this scenario because it is better suited for applications related to data lakes. It provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, perform ETL operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Cloud SQL](#)
[Cloud.google.com > What is Cloud SQL?](#)

Question #15 of 60

Question ID: 1497126

You are looking for a RESTful service for storage and access of your data on Google Cloud, as well as a way to move the data across various costing tiers based on access frequency. Which of these services meets this requirement?

- A) Cloud Filestore

✖
- B) Cloud Storage

✔
- C) Persistent Disk

✖
- D) Looker

✖

Explanation

Cloud Storage is the correct service to use in this scenario. Cloud Storage allows you to store and access data on Google Cloud using a Representational State Transfer (RESTful) service. It also provides features for security and data sharing. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets the specified criteria, such as a set period of time or when a new version of the data is created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

Persistent Disk is not the correct choice for this scenario. You would use Persistent Disk when you need a block storage service that is durable and offers high performance. Persistent Disk has storage capability utilizing Solid State Drives (SSDs) and Hard Disk Drives (HDDs), which you can attach to an instance. This instance can run in Google Kubernetes Engine (GKE) or Compute Engine.

Cloud Filestore is not the correct choice for this scenario. You would use Cloud Filestore when you need a fully managed and highly available file service that has the capability of storing persistent data for shared workloads. You would use Cloud Filestore for Enterprise-grade data applications that need storage with durability and persistence, which can be accessed using a Network File System (NFS). You can also use Cloud Filestore for storage systems that need a file system that is Portable Operating System Interface (POSIX) compliant.

Looker is not the correct choice for this scenario. You would use Looker when you need a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It can also be embedded into other applications to provide embedded analytics. You can use Looker to create alerts and proactively send timely insights that relate to various business data. It also has dashboards that are interactive, which allow you to analyze data in real-time through a trusted data model.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Cloud Storage](#)

Question #16 of 60

Question ID: 1497128

What will an organization need to do to reach the transformational phase in its AI adoption journey?

- A) Build and deploy ML models from a single platform with cross-organizational accessibility**
- B) Create a Centre of Excellence (COE) containing data scientists for creating ML solutions for specific scenarios**
- C) Deploy customized ML models into production for addressing different use cases
- D) Outsource complex ML problems and create a proof of concept using EDA tools and read-for-use AI tools

Explanation

A transformational phase in artificial intelligence (AI) adoption is achieved when a company builds and deploys machine learning (ML) models from a single platform with cross-organizational accessibility, and implements an analytics team to create ML solutions for specific scenarios across various business functions. This phase is characterized by AI teams and data scientists that are specific to products working inside of existing product teams and creating ML models that are product-specific. There is also a central analytics team that creates patterns and recommended practices for taking projects requiring ML forward. A common ML platform provides ML components, models, datasets, and components needed for creating AI solutions. The platform also provides computing that is serverless and scalable for data ingestion and batch processing.

Creating a Center of Excellence (COE) containing data scientists for creating ML solutions for specific scenarios and deploying customized ML models into production for addressing different use cases are not characteristics of the transformational phase. These are both features of the strategic phase where a centralized team seeks to create ML systems for specific product teams and allocate ML talent as per business priorities. The team also looks to identify which ML products can be purchased instead of building them, and the team assesses use cases for feasibility.

Outsourcing complex ML problems and creating a proof of concept using exploratory data analysis (EDA) tools and read-for-use AI tools is a feature of the tactical phase and not the transformational phase. The tactical phase is characterized by ML projects within the company that are not fully aligned with the organization's goals. At this stage, an organization is only exploring the feasibility of using AI for short-term goals and prototyping. An example of this is the use of AI systems for detecting written text, or building customer segmentation by utilizing descriptive analytics. At this stage, complex analytical problems are outsourced and ML projects within the company are done by individuals.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Services.google.com > Google Cloud's AI Adoption Framework](#)

Question #17 of 60

Question ID: 1497161

A company has a sketching app used by millions of users around the world, and now it needs to make some changes to the application's front end. Which would be the recommended course of action as per Site Reliability Engineering (SRE) best practices?

- A) Do not make the release until automation for toil is removed
- B) Do not make the release until there is a zero chance of errors
- C) Release the modification to a small subset of users for testing**
- D) Release the modification to all users to ensure coverage

Explanation

As per Site Reliability Engineering (SRE) best practices, it is recommended that the company release the modification to a small subset of users for testing. A key SRE best practice is ensuring that changes are released gradually with testing performed for the changes. Some SRE best practices for production servers include:

- Control failures – Ensure failures don't impact the functionality and allow systems to work using a known configuration and only use new data or configurations after they have been sanitized and validated by a human user.
- Make rollouts progressive – All rollouts and updates that need to be performed should be performed in stages and monitored throughout. The changes should be applied to a small section of traffic across time frames and geographies. This allows for problems to be detected and mitigated.
- Specify service level objectives (SLOs) from a user’s perspective – This involves setting measures for performance and availability using metrics that are applicable to a user. An example of this is the latency and error rates of a client application.
- Maintain a balance between error budgets and innovation – The system’s reliability and speed of performing innovation need to be in balance with error budgets, which are the acceptable failure levels of a specified service. An example is a 0.01% error budget for a service with 99.99% availability. This implies that a team can plan to add new features and innovation to a service only if the service has not already availed its error budget for the month, which includes errors and downtime.
- Performing monitoring – System monitoring for health can yield outputs that include pages, tickets, and logging, and require human intervention depending on the severity of a reported issue.
- Do blameless postmortems – This ensures that systems are improved in their design so that complete groups of possible errors that could arise are mitigated. It is done by not blaming people and focusing on making changes to technology and processes.
- Plan for capacity – This includes making plans to handle planned and unplanned outages while ensuring it does not affect a user beyond an acceptable limit. It involves performing load testing to find resource-to-capacity relationships and validating demand forecasts with actual demands until the two match consistently.
- Checking for overloads – This involves making sure that services still produce acceptable results even if overloaded so that users can continue using them. It may require certain features of a service to be halted in an overload situation, while the main service features are still made available.
- Limit operational tasks for SRE teams – This is to make sure that SRE teams do not dedicate more than half of their time to performing operational tasks (toil) that should be sent to development teams. For this, you can ensure there are regular meetings between SRE and development teams. Toil comprises work that may be manual, nonenduring, automatable, repetitive, and tactical.

You must not release the modification to all users to ensure coverage. As per SRE best practices, until a rollout or modification is tested and verified by a small set of users across various geographies and time frames, it must not be made available to the full set of users. This is to ensure any bugs or errors are mitigated before the change reaches all users,

Not making the release until there is a zero chance of errors is not a part of SRE best practices. Errors and failures are a natural part of the SRE process, but failure needs to be dealt with in a controlled manner. As per SRE best practices, you need to ensure that a steady state of the service is maintained so that any failure that occurs is contained and does not cause a catastrophic failure of the entire service.

Not making the release until automation for toil is removed is not an SRE best practice. On the contrary, a key goal of SRE is to ensure toil automation takes place. Toil automation removes the need for performing manual, repetitive work, which does not have an enduring value.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[SRE: A Collection of Best Practices for Production Services](#)

Question #18 of 60

Question ID: 1495242

Your company uses a hybrid cloud architecture with several of its applications needing to run on its on-premises data center as well as on Google Cloud. You need to recommend a service that can meet this requirement. Which service should you recommend?

- A) Anthos**
- B) App Engine**
- C) Firestore**
- D) Compute Engine**
- 
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- 
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Explanation

You should recommend Anthos. Using Anthos, you can create and manage an application and then run it on both on-premises systems and on the cloud. With Anthos, you can perform migrations straight from virtual machines (VMs), create and deploy apps using Google Kubernetes Engine (GKE), VMs, as well as landing zones that are serverless. Anthos allows you to use its platform to deploy both on-premises and cloud-native applications to create a global fleet that has consistency in its operations. Using containers and serverless services, Anthos lets you grow productivity. By maintaining security systems at every stage of the application's lifecycle, Anthos provides protection for both a software supply chain and an application.

You would not recommend App Engine. This cannot be used on-premises because it is a cloud computing platform. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning.

You would not recommend Compute Engine. This cannot be used on-premise because it is Infrastructure as a Service (IaaS). Compute Engine provides VM computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs to perform analytics and processing on a large scale using Google’s technological infrastructure.

You would not recommend Firestore. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database which is scalable.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Anthos](#)

Question #19 of 60

Question ID: 1495234

In the process of modernizing an on-premises infrastructure, a company wants to find a solution for moving its Oracle workloads onto Google Cloud while leveraging Oracle features like Real Application Clusters (RAC) and replication. Which Google Cloud service should you recommend for this?

- A) Bare Metal**
- B) VMware Engine**
- C) Compute Engine**
- D) App Engine**
- 
- 
- 
- 

Explanation

You should recommend Bare Metal. Bare Metal allows you to move Oracle workloads onto Google Cloud for a low risk step to infrastructure modernization. It allows you to use your on-premises database administrators, integrators, run books, and licensing. Google Cloud provides the necessary billing, service level agreements (SLAs), and support options for Bare Metal. Bare Metal provides a database architecture and supporting data center systems that are fully managed end-to-end. It also provides powerful hardware like Intel Cascade Lake servers, which have high Dynamic RAM (DRAM) intensity. Bare Metal also provides access to all existing Google Cloud services through a low latency networking system that is fully managed. You can also activate Oracle Data Guard for maintaining data backups on Cloud Storage for maximum cost effectiveness. Bare Metal allows you to have high availability using Oracle Real Application Clusters (RACs) on approved hardware.

You would not recommend VMware Engine. You would use VMware Engine for replatforming virtual machines (VMs) without changing their location. With VMware Engine you can move applications that are based on VMware onto Google Cloud without any changes to existing applications. It provides all the necessary infrastructure and VMware licenses that allow you to have a software-defined data center (SDDC) on Google Cloud. Using VMware Engine provides elasticity and scaling through the ability to swiftly build a private cloud using flexible management of resources and automatic scaling.

You would not recommend App Engine. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning.

You would not recommend Compute Engine. Compute Engine provides VM computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs, and it allows you to perform analytics and processing on a large scale using Google's technological infrastructure.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Bare Metal Solution for Oracle](#)

Question #20 of 60

Question ID: 1497155

Your company has recently moved to a cloud deployment and is struggling to manage its growing cloud costs. You are asked to suggest some ways to get costs under control. Which of these should you suggest? (Choose two.)

- A) Set usage quotas for resources**
- B) Create alerts for budget overruns**
- C) Duplicate data across buckets
- D) Run all VMs continuously
- E) Store all data with the standard class

Explanation

You should suggest the company set usage quotas for resources and create alerts for budget overruns. Some controls available to companies for keeping cloud costs and usage under control include budgetary alerts, usage quotas, and company permissions. Setting resource quotas and performing throttling must ensure that cost overruns do not occur due to resource utilization exceeding the specified budgets. Using Google Cloud tools, you can perform automated throttling, capping of resource consumption, and broadcast budget notification alerts on company communication channels, such as Slack.

You would not store all data with the standard class. This action can cause wasted capital due to overspending on storage. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

You would not run all virtual machines (VMs) continuously. Using auto start and stop for VMs can help reduce Compute Engine's operating costs. By deactivating VMs during non-business hours rather than allowing VMs to run continuously, you can gain significant cost savings. You can use a serverless system to automate the process of VM shutdowns by using Cloud Pub/Sub, Cloud Scheduler, and Cloud Functions.

You would not duplicate data across buckets. Duplicating data causes a waste of storage and capital. Some ways you can prevent creating duplicate data is by using multi-region buckets and setting object versioning on Cloud Storage correctly. By specifying an object versioning policy, you can maintain a suitable number of copies of data, and you can also use the bucket lock version to protect from accidental deletion of objects.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Question #21 of 60

Question ID: 1497132

Your company is using an ML model or making sales predictions based on customer shopping preferences. So far, the predictions have not been good. Which of these could be the cause of poor prediction quality in the ML model?

- A) Placing data validation at the start of the ML pipeline
- B) Having transformational maturity in ML applications
- C) Usage of skewed data**
- D) Performing continuous training

Explanation

Skewed and anomalous data are primary cause of bad predictions made by machine learning (ML) models. ML needs to have some quality control measures implemented for the best results. These measures include keeping processes to evaluate models after training and to validate them pre-deployment. Additionally, there needs to be logging and monitoring systems in place so that the system's health can be checked, and notifications can be issued in the event of any anomaly. When continuous integration/continuous delivery (CI/CD) is applied to ML, it involves validation of ML models, data schemas, and code as part of CI, and deployment of ML models and continuous testing (CT) pipelines as part of CD. To create a production system for ML, you would use ML Metadata, Container Registry Cloud Build, and Model Registry services.

Continuous training (CT) is not a cause for bad predictions made by ML models. CT helps validate all the components and code of ML systems and checks existing models and schemas. The purpose of CT is to help ML models adapt to changes in data before the models are deployed. This is done automatically. To automate CT reliably, you must place validation and quality checks for ML training into the existing ML pipeline. Data validation measures need to be placed at the beginning of the pipeline and model validation needs to be placed at the end. Data validation is used to verify the data types and schemas for the fresh data, which is used to retrain an existing ML model. Model validation checks the performance of an ML model's predictions and ensures that the performance is better than its predecessor.

Transformational maturity in ML applications is not a cause for bad predictions made by ML models. Transformational maturity has systems in place to ensure optimum performance of ML models. Transformational maturity is characterized by the creation of systems for the online processing of real-time events and the creation of a unified ML platform utilized by ML specialists across multiple teams. There is also a provision of compute resources, recommended practices, tools, and services needed by ML teams to complete ML projects. This maturity level also sees the use of tools to accelerate ML computing like Cloud Tensor Processing Units (TPUs) and Graphics Processing Units (GPUs) and the use of templates for data processing, which are driven by metadata so workflows can be deployed without the need for coding. Transformational maturity involves orchestration and automation of ML and data pipelines end-to-end.

Placing data validation at the start of the ML pipeline is not a cause for bad predictions made by ML models. Data validation needs to be kept at the start of the pipeline. This is because the data types and schemas used for model training are correct. Without this step, the data fed to an ML model may not be accurate, which leads to poor results in ML model predictions. Additionally, you need to place the model validation process at the end of the pipeline so that you can verify whether or not the trained model is meeting prediction expectations better than the current ML model in use.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Services.google.com > Google Cloud's AI Adoption Framework](#)

Question #22 of 60

Question ID: 1497156

Your company's C-Suite has asked you to take over the task of directing the management of its ongoing Google Cloud expenses. Which of these actions must a company perform to achieve better financial governance of its cloud deployment?

- A) Implement throttling on cloud resources exceeding quotas**

✔
- B) Ensure the on-premises data center matches the cloud for hardware**

✘
- C) Keep teams siloed for better individual focus on their work**

✘
- D) Allow all users across the company to provision and access resources**

✘

Explanation

A company must implement throttling on cloud resources that exceed quotas to achieve better financial governance of a cloud deployment. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budgets. A company can keep various controls when considering a public cloud deployment. These can help ensure better governance over cloud expenditure.

- Specify permissions for cloud resource usage using the principle of least privilege
- Maintain control of cloud resource usage by implementing resource quotas
- Control cloud expenditure using budgetary action automation and notifications

You would not allow all users across the company to provision and access resources. This can cause improper use of cloud resources that leads to wasted capital as well as security issues. Instead, you would opt to use the principle of least privilege for specifying access permissions. The impact of the least privilege principle in access management is that a user only has access to the cloud resources necessary for task completion. This ensures that a user cannot access resources that are outside their scope of work, thus increasing privacy and security on the cloud. This access model ensures that users with malicious intent cannot cause damage to resources outside of their privileges.

You would not ensure the on-premises data center matches the cloud for hardware. The on-premises data center is independent of the cloud deployment in its infrastructure. The purpose of using a cloud deployment is to have a cost-optimized way of scaling computing and storage resources to match business demands without needing to take on the capital expenditure associated with procuring the same level of infrastructure required for a data center.

You would not keep teams siloed for better individual focus on their work. By incorporating a partnership that is implemented across various teams, you will gain greater visibility, which is a necessity for efficient cost management. Reducing budget overruns and unauthorized resource use can be accomplished with more collaboration between IT and finance teams.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Question #23 of 60

Question ID: 1495929

Your company has an on-premises data center that it uses to provide data processing services to customers. Due to the nature of the legacy system, there have been some business disruptions to the current users of specific services due to network traffic issues, anomalies, and malicious bot attacks.

What would be the best way to solve this issue?

- A) Move apps to the cloud using VMware engine**

✘
- B) Use API management with Apigee**

✔
- C) Create Kubernetes clusters with GKE**

✘
- D) Build a document database with Firestore**

✘

Explanation

You should use application programming interface (API) management with Apigee to solve this issue. API management with Apigee enables you to perform monitoring and analysis of API traffic to help mitigate key problems related to performance. It also allows you to build a layered defense system that includes distributed denial of service (DDoS) attack protection, bot handling, API guarding, as well as web application firewalls (WAFs). Apigee lets you monitor APIs and ensure your APIs are delivering expected performance and maintaining high availability for customers and developers. You can also identify any API issues and diagnose them easily using the monitoring systems for Apigee. API management with Apigee allows you to perform routing based on context, transform payloads, and perform API facades that are user specific. You can start by routing a small portion of the users to the modernized version of the service that uses APIs, and then based on the user feedback and experience, you can route all the users to the API-based modern system.

You would not create Kubernetes clusters with Google Kubernetes Engine (GKE) for this scenario. GKE is not used for API management, which is what this scenario needs. GKE allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes enabling you to specify CPU, RAM, storage, GPUs, and network configurations along with an Autopilot option for situations where there is not enough operational and developer support available.

You would not build a document database with Firestore. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not move apps to the cloud using VMware engine. VMware Engine can be used to move applications that are based on VMware onto Google Cloud without any changes to existing applications. It provides all the necessary infrastructure and VMware licenses that allow you to have a software-defined data center (SDDC) on Google Cloud.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Question #24 of 60

Question ID: 1497150

Your company has various teams and departments, all of which concurrently use resources on a Google Cloud deployment. How should you implement financial governance on Google Cloud to allow your company to achieve more accountability in its cloud operations?

- A) Using cloud resource labels and hierarchies**
- Keeping a flat hierarchy with a single organizational level
- Managing inventory for on-premises systems
- Ensuring all resources are owned by a single user account
- ✔

✖

✖

✖

Explanation

To achieve greater accountability in your cloud deployment for financial governance, you should use cloud resource labels and hierarchies. Using resource hierarchies simplifies the allocation of costs to specific departments and ensures accountability. This action also provides insights into the return on investment (ROI) for your cloud deployment. A resource hierarchy enables you to create a structure for your Cloud account based on your specific business needs. The hierarchy can be described below:

- Level 1 (topmost) – Organization: The Company
- Level 2 – Folders: Departments, shared infrastructure, teams within departments, and products within teams
- Level 3 – Projects: Development, testing, and production using Google Cloud Platform (GCP) for projects
- Level 4 – Resources: Compute Engine instances, Cloud Storage buckets, and App Engine services

Additionally, resource labels can be used to group several resources together and to identify the teams that own particular resources. As an example, resource labels can be App:UI, Env:Prod, and Dept:Testing, which indicate that a resource is for an app called UI in the production environment used by the testing department.

A company needs various financial governance tools for easy cost management that ensures clarity, accountability, and control. Google Cloud financial governance helps a company control costs by providing various tools and services.

- Clarity – This is achieved using billing reports from the Google Cloud Platform (GCP) console, billing application programming interfaces (APIs), and intelligent recommendations.
- Accountability – This is achieved using resource hierarchies and resource labels.
- Control – This is achieved using organizational policies and cost management controls.

Managing inventory for on-premises systems is not a way of implementing accountability on a cloud deployment. An on-premises system and its inventory are independent of cloud resources. The financial management of on-premises systems is not related to the financial governance systems that need to be implemented to manage costs on a cloud deployment.

Ensuring a single user account owns all resources is not a way of implementing accountability on a cloud deployment because a cloud deployment for a company can have several different departments using the system. For this, there needs to be a way to have transparency so the company knows the exact users and departments that utilize specific cloud resources.

Keeping a flat hierarchy with a single organizational level is not a way of implementing accountability on cloud deployment. Keeping a multi-level hierarchy that reflects the operating structure of an organization is critical for providing accurate financial accountability in the cloud deployment. By having various levels of scale, costs can be allocated correctly and managed efficiently based on the specific departments and projects that utilize the resources. With a flat organizational hierarchy, it would be impossible to have accountability for resource utilization and the associated costs of specific departments and projects.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Question #25 of 60

Question ID: 1495909

You are in the process of designing a data warehousing solution for a company. Which of these would be the key characteristics of the data warehouse? (Choose two.)

- A) Contains unstructured data, which is accessed using code**
- B) Provides non-persistent storage**
- Answers known queries by accessing known data
- Provides a way to access unknown data
- Allows SQL for data access and manipulation
- ✖

✖

✔

✖

✔

Explanation

The following options are the correct characteristics of a data warehouse:

- Answering known queries by accessing known data
- Allowing SQL for data access and manipulation

A data warehouse is a massive repository that contains a company's data gathered from various different sources. This data includes current data as well as data that is historical. A data warehouse is used for business intelligence (BI) activities, such as complex analytics, which can lead to making informed management decisions.

The following options are not characteristics of a data warehouse but of a data lake:

- Unstructured data, which is accessed using code
- Providing a way to access unknown data

A data lake allows you to store both structured, semi-structured, and unstructured data from multiple sources in a single place. This data can come from a variety of sources, including multimedia, social media, hard disks, servers, Internet of Things (IoT) devices, clickstreams, databases, and transactional information.

A data warehouse is not used for providing non-persistent storage but is a storage repository for the organization's data. The sources of data for a data warehouse include customer relationship management (CRM) systems, automation systems for marketing, applications for transactions, and log files from applications. A data warehouse can be used for performing analysis in an ad hoc manner or for creating custom reports.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Services.google.com > Build a modern, unified analytics data platform > Data warehouse or data lake?](#)

Question #26 of 60

Question ID: 1495257

A shopping company has several self-service terminals at various locations around the country that its customers can use to make payments and get information on discounts. Using this deployed hardware, the company also wants to leverage a new virtual application that will service customer requests. Why will the company need to use APIs with this system?

- A)** To perform data lifecycle management using object storage
- B)** To allow customers to access the Internet from the terminals
- C) To allow the new application to communicate with the check-out hardware**
- D)** To change key hardware components in the terminals

Explanation
The company will need to use application programming interfaces (APIs) to allow the new application to communicate with the check-out hardware. With Cloud APIs, you can automate workflows using whichever programming language you prefer and use Representational State Transfer (REST) calls and client libraries as required. APIs can facilitate communication between legacy systems like check-out hardware and modern virtual customer service applications.

Performing data lifecycle management using object storage is not related to this scenario. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria, such as a period of time passing, or a new version of the data being created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

Changing key hardware components in the terminals cannot be done using APIs. Changes to hardware components can be done by technicians and the department that handles hardware control and maintenance. APIs are a way for applications to communicate with each other and share services.

Allowing customers to access the Internet from the terminals is not the reason why you need to use APIs with this system. Customers should not be allowed to use the terminals for Internet access as it can pose a significant security risk to the system. A self-service terminal needs to be protected and secured from Internet access for safety.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Cloud APIs](#)

Question #27 of 60

Question ID: 1495181

You work for a company that is exploring how Google Cloud can allow it to apply its data in new ways. The company wants its applications to be responsive to changing user needs, which can be serviced on demand. What aspect of Google Cloud would be useful in this scenario?

- A)** Storage of data as archives for access over a lengthy period
- B) Ingestion and analysis of user data performed in real-time**
- C)** Validation and cleaning of data done automatically
- D)** Removal of overhead related to managing data access

Explanation

Ingestion and analysis of user data performed in real-time is a distinct advantage of using Google Cloud that is relevant to this scenario. To be responsive to changing user needs and preferences, a real-time system is required to collect user data. Gathering and analyzing user data in real-time is done easily on a cloud deployment. Such real-time systems allow applications to make necessary immediate changes to address various user requirements dynamically based on ingested real-time information.

Storage of data as archives for access over a lengthy period is not a relevant advantage in this scenario. Archiving data is a benefit of using the cloud, but it is for data that does not need to be accessed immediately and is not suitable for real-time processing needs. Archiving is a way to keep low-cost storage for infrequently accessed data and meet regulatory and compliance requirements. Archiving using the cloud is not a way to implement real-time user interaction and data requirements.

Validation and cleaning of data done automatically is not a relevant advantage in this scenario. Automatic validation and cleaning of data are useful for applications that need quick and efficient data processing, such as machine learning (ML) applications that better serve users in an efficient and automated manner. The company takes responsibility for the quality and validity of the stored data. This does not represent the kind of real-time ingestion and analysis that is needed in this scenario.

Removal of overhead related to managing data access is not a relevant advantage in this scenario, which requires real-time data processing. Having a centralized Identity and Access Management (IAM) system using user access policies is a key advantage of using a cloud system. When an organization moves to a cloud deployment, it needs to take full responsibility for the access management of its data. That is not the responsibility of the cloud service provider (CSP), but the customer. The CSP looks after the security of the infrastructure that supports the cloud but not of data stored in the cloud by the customer.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Describe the role of data in digital transformation and the importance of a data-driven culture

References:
[Cloud.google.com > Google's guide to building a data-driven culture](#)

Question #28 of 60

Question ID: 1495930

A company with an on-premises data center needs to understand the benefits of adopting an API-based system. Which of these tasks using APIs can modernize an existing legacy on-premises system?

- A) Ensuring all teams perform tasks at the same pace

✖
- B) Splitting applications into individual business services

✔
- C) Reducing automation to give developers more control

✖
- D) Creating a reliable monolithic application for serving users

✖

Explanation

By using application programming interfaces (APIs), you can split existing applications into individual business services. The process of functional decomposition involves identifying individual components and their related systems inside of existing workloads. Then, you can create a decoupled architecture where workloads are broken into smaller parts and scaled independently. This can help you provide higher availability of services and protection from failures. With APIs for various business services, multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Reducing automation to give developers more control is not a benefit of using APIs. One of the main advantages of using APIs is to provide automation that reduces the time users spend doing tasks by having more tasks performed by computers. This boosts productivity and also can provide more accountability with tracking systems that record API calls.

Creating a reliable monolithic application for serving users is not a benefit of using APIs. APIs help move away from a monolithic architecture for application development to a microservices-based architecture. This provides greater stability to the system and ensures higher availability as individual components of the system can run independently, and there is no dependency between them. This ensures that if one component fails, then the other application components are not affected.

Ensuring all teams perform tasks at the same pace is not a benefit of using APIs. APIs can be used for various business services, and multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Question #29 of 60

Question ID: 1497159

You work for a multinational company that has recently performed a cloud adoption process. The company needs you to find ways to have more transparency into cloud expenses for better management. What cost management strategy should you use in this scenario?

- A) Delegate responsibility of cloud budgeting to one stakeholder

✖
- B) Identify any cloud cost overruns to the company's error budget

✖
- C) Increase scrutiny of on-premises technology and services

✖
- D) Facilitate collaboration between financial, IT, and marketing teams

✔

Explanation

You should facilitate collaboration between financial, IT, and marketing teams. This is necessary to ensure visibility and transparency as part of cost management. By incorporating a partnership that is implemented across various teams, you will gain greater visibility, which is necessary for efficient cost management. Reducing budget overruns and unauthorized resource use can be done with more collaboration between IT and finance teams.

You would not delegate the responsibility of cloud budgeting to one stakeholder. Keeping a single person in charge of managing cloud costs cannot work for a large organization due the scale and complexity involved. An individual can look after cloud cost management only for very small companies. Large companies will require cross-team collaboration for effective cost management.

You would not identify any cloud cost overruns to the company's error budget. An error budget is not related to cloud expenses. It is the variation between the service level agreement (SLA) and service level objective (SLO). The error budget is the measure of the system's possible downtime, which is the difference between the SLO and the total availability figure of 100%. The SLO is kept at a higher value than the SLA to ensure early warnings of possible availability issues.

You would not increase scrutiny of on-premises technology and services. The technology and services operating inside on-premises systems do not have an effect on the costs of running a cloud deployment. The financial management of on-premises systems is not related to the financial governance systems that need to be implemented to manage costs on a cloud deployment.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Your company has embraced a modern application architecture by dividing existing systems into microservices and setting up services management for them. Now, what should you use to manage microservices as APIs?

- A) BigQuery

B) Dataprep

C) Apigee

D) Firestore
- ✖

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Explanation

You should use Apigee for managing microservices as application programming interfaces (APIs). Apigee API Management works in tandem with Envoy and Lstio by enabling API management within a microservices stack. Lstio is an open source system for managing and connecting microservices securely. Apigee is used to build, secure, scale, and analyze APIs at any location with support for the complete API lifecycle. It provides strong systems of API control and visibility. With Apigee, you can increase the value of your digital assets by monetizing your API-based products.

You would not use Firestore for this scenario. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use and has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

You would not use Dataprep for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for purposes of ML and analysis. It is an intelligent data service and is serverless. This removes the need for users to deploy or manage infrastructure. It also suggests data transformations that are ideal each time a user interface (UI) input is detected, and this removes the need for a user to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. This helps a user avoid having to assess the quality of data and proceed directly to exploratory data analysis.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Question #31 of 60

Your company has multiple teams that have hundreds of employees in them. You need a centralized way to manage user access to cloud resources. Which of these systems would you need to use to authorize which users can access specific Google Cloud resources?

- A) Billing APIs

B) Secret Manager

C) Cloud IAM

D) Encryption at rest
- ✖

✖

✔

✖

Explanation

You would use Cloud Identity and Access Management (IAM) for this scenario. Cloud IAM enables you to manage cloud resources centrally with granular access control and visibility. With IAM, you can grant access at the resource level, such as providing the role of Subscriber to a user for a Pub/Sub topic. This way, IAM allows you to grant roles in addition to Owner, Editor, and Viewer roles. IAM also provides Recommender that can enable you to detect overly permissive access policies and rightsize them. With IAM, resource access is controlled using contextual attributes, such as IP address or resource type. You can perform IAM policy management and creation using the Google Cloud Console, programmatically using IAM methods, or by command-line access using the gcloud tool. Google Accounts are also supported by IAM. You can utilize Cloud Identity with IAM to synchronize user accounts across multiple projects. IAM also provides workforce identity federation that enables you to perform authentication and authorization of a workforce using external identity providers (IdPs). IAM can provide a full audit history that includes granting, removal, and delegation of permissions for compliance needs.

You would not use Secret Manager for this scenario. You use Secret Manager for storing sensitive data like passwords, application programming interface (API) keys, and certificates securely. With Secret Manager, you have a single source of truth for accessing, managing, and auditing secrets on a Google Cloud deployment. Secret Manager provides lifecycle management with versioning for pinning requests to the newest versions of secrets, as well as the ability to use Cloud Functions to automate the process of rotating secrets. It also allows you to select regions for the replication of your secrets data. Secret Manager is integrated with Cloud IAM for controlling access to secrets.

You would not use encryption at rest in this scenario. Encryption at rest of stored data is a way of implementing security for data stored on the cloud infrastructure. Google utilizes multiple layers of encryption for the protection of data at rest. Google's storage infrastructure encrypts all user data by default before it is stored on physical storage. This way, encryption is performed by Google Cloud at the storage or application layer of the cloud infrastructure. Google has several storage services like Colossus and Cloud Spanner, as well as a key management service.

You would not use billing APIs for this scenario. You can use billing APIs to access various tools created by other vendors to manage project bills programmatically and obtain data on all stock keeping units (SKUs) in their repository and related metadata. This is a way of ensuring clarity for financial governance on a cloud deployment. Other methods include using intelligence recommendations for performing rightsizing of computing resources. This helps ensure that costs are minimized by ensuring that virtual machine (VM) usage is optimized.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Cloud Identity and Access Management \(IAM\)](#)

Question #32 of 60

You want to create a cloud-based testing environment that duplicates various different hardware setups so that the testing team can perform various application tests within this isolated environment. Which of these technologies should you use for this requirement?

- A) Physical computers

B) Serverless data warehouses

C) Virtual machines

D) Containers
- ✖

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Explanation

This is a use case for virtual machines (VMs). A VM is a virtual computer that runs operating systems and applications. Multiple VMs can run on a single physical computer and access various resources from a hypervisor. A hypervisor is software that creates an abstraction of all the computing resources of the physical computer and provides these resources to VMs running on the computer as required. The VM architecture provides the underlying framework in use for modern cloud computing. Compute Engine is Google's VM service, which offers VMs that can be customized. Some uses of VMs in enterprises include:

- Server consolidation – VMs can act as servers to host other VMs and so reduce sprawl by consolidating multiple computing systems into a single physical machine.
- Migration of workloads – VMs are flexible and portable, which enable swift workload migration for companies.
- Test environment creation – VMs can provide an isolated system that can be used for testing without affecting the surrounding systems.
- Disaster recovery – You can duplicate an existing system as a VM on the cloud, thus adding an additional layer of security to the system.
- DevOps – VMs can be activated, deactivated, and migrated smoothly, which facilitates flexible app development.
- Hybrid systems – VMs help you flexibly integrate existing on-premises legacy systems with a cloud environment to create a hybrid computing environment.

You would not use containers for this scenario. Containers are better suited for creating applications that need to be compartmentalized and made portable across multiple platforms. By compartmentalizing an application using containers, you can then change parts of the application in isolation from the other parts. Containers are software packages that contain application code as well as dependencies, such as runtime environments and libraries, that are required for running an application. This allows containers to virtualize an OS and allow applications to be made portable across multiple systems.

You would not use serverless data warehouses for this scenario. A serverless system like BigQuery would be suitable for a requirement where there needs to be very high levels of scalability for working with very large quantities of data. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use. It also has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

You would not use physical computers for this scenario. This is because procuring physical computers for creating testing environments is not as economical, feasible, or flexible as using a VM based system. A single physical computer has one hardware configuration with a specific CPU, RAM, and storage setup. Using several different VMs, you can have several hardware setups, which makes the process of testing an application across multiple platforms much simpler.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > What is a Virtual Machine?](#)

Question #33 of 60

Question ID: 1495221

A game development company has been developing games from a legacy on-premises data center and is now considering making a digital transformation. How would the company benefit from modernizing its infrastructure with cloud technology?

- A) Making significant investments in new emergent technologies**
- B) Ensuring all resources are paid for regardless of usage**
- C) Delegating application security to the cloud service provider**
- D) Massive scalability by using serverless technology**
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Explanation

Using cloud-based technologies like serverless computing, a company can achieve massive levels of scalability, which may not be feasible using traditional legacy systems. For example, BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

Delegating application security to the cloud service provider (CSP) is not a benefit of using cloud technology. Security of applications and all the data stored in the cloud is the responsibility of the customer and not the CSP. The CSP is responsible for the security of its data centers and of the cloud.

Paying for all resources, regardless of usage, is not a benefit of using cloud technology. Cloud computing ensures that companies only pay for resources consumed. This ensures that data centers do not need to be provisioned with more capacity to handle spikes in usage or organizational growth, and allows a company to have its staff work on more strategic tasks that help grow the business.

Making significant investments in new emergent technologies is not a benefit of using cloud technology. CSPs have access to the latest technologies, which customers can avail of as services. This removes the risk of the company having to make large investments into new technologies, which may become obsolete in time.

Some distinct benefits of cloud technology are outlined below:

- Security – Through a shared responsibility model and best practices, you can ensure greater safety of your operations using Google Cloud.
- Reliability – This helps you ensure your workloads can perform correctly, handle failures, and automatically fix any issues that may arise. This also covers data backups, fault isolation, and disaster scenario preparation.
- High Availability – This ensures that your applications are always ready to serve customers swiftly with almost no downtime. A measure of availability is five 9s, which corresponds to a system being available 99.999% of the time.
- Elasticity – This is the ability to scale your computing resources to match consumer demand without having to perform resource provisioning manually or guess future capacity needs.
- Agility – This is the ability to swiftly deploy new applications by removing unnecessary operational overhead so that the development team can focus on creating and innovating products.
- Pay-as-you-go (PAYG) pricing – This is a flexible payment option where a cloud consumer only pays for resources consumed.
- Scalability – This is using the right technology to easily make your application meet growing demands on a massive scale using serverless technologies.
- Global Reach – Using edge locations and data centers around the world, Google Cloud's global infrastructure can improve user experience with low latency for end users, regardless of their location on the globe.
- Economy of scale – This is having lower PAYG pricing due to accumulated cloud usage from a very large number of cloud customers.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > What is Cloud Computing? > What are the benefits of cloud computing?](#)

Question #34 of 60

Question ID: 1497130

How would you implement transformational maturity for an organization using cloud-native ML technology?

- A) Acquire dedicated cloud computing resources for selected data scientists in the company**
- B) Perform queries for analytics in an ad hoc way using BigQuery**
- C) Perform exploratory analysis of data offline using datasets that are small**
- D) Use ML to create pipelines for stream analytics and complex event processing**
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Explanation

Using machine learning (ML) to create pipelines for stream analytics and complex event processing are characteristic of transformational maturity. Transformational maturity is characterized by the creation of systems for online processing of real-time events and the creation of a unified ML platform that is utilized by ML specialists across multiple teams. There is also a provision of compute resources, recommended practices, tools, and services needed by ML teams to complete ML projects. This maturity level also includes the use of tools to accelerate ML computing, such as Cloud Tensor Processing Units (TPUs) and Graphics Processing Units (GPUs), as well as the use of templates for data processing, which is driven by metadata so workflows can be deployed without the need for coding. Transformational maturity involves orchestration and automation of ML and data pipelines end-to-end.

Performing queries for analytics in an ad hoc way using BigQuery is characteristic of strategic maturity. This maturity level includes the creation of a cloud data warehouse that is enterprise-level, fully managed, and used to perform analytical queries ad hoc. The data for the data warehouse is ingested from multiple sources and used by data scientists within the company for analytics. The process of ingestion and transformation of data is done using serverless and fault tolerant tools, which have parallel processing and autoscaling capabilities. These tools work with both big batches as well as Extract, Transform, and Load/Extract, Load, and Transform (ETL/ELT) workloads for the processes of populating data warehouses and creating datasets for ML. Some of these tools include Cloud Data Fusion, Dataflow, and Dataproc, and they work with both structured and unstructured data for gathering key insights.

Acquiring dedicated cloud computing resources for selected data scientists in the company and performing exploratory analysis of data offline using datasets that are small are both characteristics of tactical maturity. This maturity level is characterized by the use of limited computing resources like data science virtual machines (VMs) for local data processing and training ML models. The tactical maturity level also involves the exploration of AI possibilities through use cases for proof of value (POV) demonstrations. This maturity level sees controlled costs for ML exploration with hardware and compute resources that are dedicated for use by selected data scientists. The exploratory analysis of data is done using ready-for-use AI systems and tools for data wrangling.

Objective:

Innovating with Data and Google Cloud

Sub-Objective:

Identify Google Cloud's solutions for machine learning and AI

References:

[Services.google.com](#) > [Google Cloud's AI Adoption Framework](#)

Question #35 of 60

Question ID: 1497145

You are in the process of creating a working environment for developers in your various machine learning teams. Which of these options would you use for a development environment that is based on a Jupyter notebook and allows for data processing using a Dataproc cluster?

- A) Dialogflow

B) CCAI

C) Vertex AI Workbench

D) Firestore
- ✖

✖

✔

✖

Explanation

You would use Vertex Artificial Intelligence (AI) Workbench for this scenario. This provides a development environment that is based on the Jupyter notebook system and can be utilized for the entire workflow you use for your data science tasks. With Vertex AI Workbench, you can work with data in Dataproc clusters, perform model training, and propagate the results from within the interface of JupyterLab. Vertex AI also contains data labeling jobs that you can use to perform data labeling for datasets that will be used for model training. The Vertex AI Feature Store provides a repository that is fully managed for ingestion, serving, and sharing feature values of ML inside of a company.

Dialogflow is not the appropriate choice for this scenario. Dialogflow provides a suite for the development of conversational apps with voice and text capabilities, which include voicebots as well as chatbots. Dialogflow provides cross-platform connectivity to existing apps for the Internet of Things (IoT), iOS, or Android, and for telephony platforms, including Cisco, Genesys, and Avaya. It also can be used for digital platforms, including Slack, Facebook Messenger, and Google. Dialogflow Essentials (ES) is the standard edition, and Dialogflow Customer Experience Edition (CX) is the advanced edition with features like flow visualizations that are interactive and advanced AI features.

Firestore is not the appropriate choice for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT) that can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

Contact Center Artificial Intelligence (CCAI) is not the appropriate choice for this scenario. CCAI is used for enhancing the customer service experience for your customers in existing contact centers by leveraging AI technology. CCAI includes Dialogflow CX, Dialogflow Essentials, Speaker ID, Text-to-Speech, and Speech-to-Text. The CCAI Platform is a Contact Center as a Service (CCaaS) platform that is AI enabled and created on Google Cloud natively. It is built to be used with Customer Relationship Management (CRM) systems and provides a unified source of data for customer experiences. Contact Center AI Insights enables customers to perform value extraction from their data in contact centers. They can use Contact Center AI Insights to search for key information and perform necessary actions based on that data using a console. Customers can also use Contact Center AI Insights for extraction of topics, key areas, and sentiments from data using advanced analytics.

Objective:

Innovating with Data and Google Cloud

Sub-Objective:

Identify Google Cloud's solutions for machine learning and AI

References:

[Cloud.google.com](#) > [Introduction to Vertex AI](#)

Question #36 of 60

Question ID: 1495293

In an effort to have greater control over the performance of applications running on the Cloud system, a company's operations team wants to investigate the workings of the applications. What Google Cloud service should your operations team use to get a better understanding of what occurs when requests traverse through the system?

- A) Cloud Trace

B) Cloud Armor

C) Cloud Logging

D) Cloud Monitoring
- ✔

✖

✖

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Explanation

Your operations team should use Cloud Trace for this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights into its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app. Application Performance Management (APM) is a combination of several tools that ensure optimal performance of applications. APM integrates tools to monitor and troubleshoot applications, which include Cloud Monitoring and Cloud Logging. It also includes latency and cost reduction systems like Cloud Trace and Cloud Profiler. This boosts the efficiency of operations.

You would not use Cloud Monitoring for this scenario. Cloud Monitoring can be used to monitor the complete infrastructure of a cloud deployment. It provides a view into the uptime, health, and performance of cloud systems. For this, Cloud Monitoring gathers relevant metadata, events, and metrics from sources that include uptime probes, services, app instrumentation, alerts, notifications, and other components of applications.

You would not use Cloud Logging for this scenario. Cloud Logging is a log management service that is fully managed. It allows you to store, search, and analyze exabytes of log data, as well as set up alerts and notifications. Cloud Logging has Logs Explorer, which lets you use flexible queries for searching, sorting, and analyzing logs. You can also utilize log buckets for storing and processing log data based on regional compliance requirements.

You would not use Cloud Armor for this scenario. Cloud Armor is a service that protects websites and software applications from threats like web attacks and distributed denial of service (DDoS) attacks. It offers adaptive protection by detecting and mitigating large numbers of Layer 7 DDoS incidents using a locally trained machine learning (ML) system.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Trace](#)

Question #37 of 60

Question ID: 1495238

You are creating a plan for digitally transforming workloads for your company. What would be the purpose of using cloud-native applications during application modernization?

- A)** Partitioning workloads for failure protection
- B)** Consolidating a single unified backend for redundancy
- C)** Avoiding cost overruns caused by application decoupling
- D) Using APIs for ensuring all business teams work at the same pace**

Explanation

Using cloud-native applications allows for partitioning workloads for failure protection. The process of functional decomposition involves identifying individual components and their related systems inside of existing workloads. Then, you can create a decoupled architecture where workloads are broken into smaller parts and scaled independently. This can help you provide higher availability of services as well as protection from failures by maintaining several read replicas of a service within a system that is distributed.

Consolidating a single unified backend for redundancy is not a purpose of application modernization. You would instead spread backends and other services over several availability zones (AZs) and use read replicas and backups to ensure that the system is not affected by failures. You may need to maintain certain services in a specific geographical location for compliance, and you may need to distribute other services globally for low latency content distribution to users.

Avoiding cost overruns caused by application decoupling is not a purpose of application modernization. Performing decoupling of applications as part of application modernization ensures that services that scale up or down more frequently are separated from other services that are more stable. This way, you cut back on costs associated with scaling up of services that are not required or are unrelated.

Using application programming interfaces (APIs) to ensure all business teams work at the same pace is not a purpose of application modernization. You would use APIs to perform application decoupling for existing business services. This is a form of functional decomposition that breaks down an existing legacy or “monolith” application into independent applications. With APIs for various business services, multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Question #38 of 60

Question ID: 1495285

A software development company is using Google Cloud for its development environment. It is now considering using DevOps. What is the purpose of incorporating DevOps into a production environment? (Choose two.)

- A) Make deployments quicker**
- B)** Lessen the need for monitoring tools
- C) Increase software stability**
- D)** Increase in-person processes
- E)** Perform less frequent testing

Explanation

Increasing software stability and making deployments quicker are both goals of using DevOps. DevOps is an initiative that is geared toward increasing the speed of delivering software, making services more reliable, and creating ownership that is shared among all stakeholders. DevOps principles are used to enforce security best practices from the beginning of each development project. Google's DevOps Research and Assessment (DORA) team has found certain key capabilities that an organization can leverage for improved performance and faster software delivery. These capabilities are related to the company's culture, measurement systems, technology, and processes.

Increasing in-person processes is not the purpose of DevOps. Using automation is a key feature of the DevOps philosophy, which seeks to create an automated model for application development that results in more error-free operations. The automated approach also helps ensure that changes can be rolled back, if required, and system stability is ensured using safe configurations.

Performing less frequent testing is not the purpose of DevOps. A key DevOps practice is continuous integration and continuous delivery (CI/CD), which specifies that a controlled and systematic approach needs to be used for making changes to ensure system stability and avoid configuration drift. A public cloud system offers features like CI/CD for rapid and automated development and updates for applications. You would first use a simple CI model and then employ CD. After some time, you would apply the CI/CD model to all existing services. Then you can add additional stages to your CI/CD pipeline gradually. Continuous testing (CT) is a part of CD, which ensures testing occurs regularly and in sync with delivery.

Lessening the need for monitoring tools is not the purpose of DevOps. The system health and reliability need to be monitored and measured as part of the DevOps philosophy. For this, various monitoring tools need to be used. Monitoring helps understand the performance of applications and map resource consumption to specific code areas of the application for further optimization.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Cloud.google.com > DevOps](#)

Question #39 of 60

Question ID: 1495187

An e-commerce company has consulted with you to improve its user shopping experience. You seek an automated service to provision processing resources needed to analyze clickstreams from users and create optimizations for their shopping experience through promotions and real-time prices. Which of these services should you use for this?

- A) Cloud Trace**
- B) Cloud Spanner**
- C) Dataprep**
- D) Dataflow**
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-
-
-

Explanation
You should use Dataflow for your streaming analytics processing needs. Dataflow provides an economical, swift, and serverless system for stream and batch processing work. It is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources. Along with Dataflow, you can utilize Pub/Sub and BigQuery for working with stream analytics, which involves ingestion, processing, and analyzing changing amounts of data produced in real-time for organizational insights and initiatives.

You would not use Dataprep for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for the purpose of machine learning (ML) and analysis. It is an intelligent data service and is serverless. This removes the need for users to deploy or manage infrastructure. It also suggests data transformations that are ideal each time a user interface (UI) input is detected, and removes the need for a user to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. This helps a user avoid having to assess the quality of data and proceed directly to exploratory data analysis.

You would not use Cloud Trace in this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

You would not use Cloud Spanner for this application. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner to provision a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Dataflow](#)

Question #40 of 60

Question ID: 1495904

You have recently created a cross-functional cloud team for furthering cloud adoption in the company. How would you evolve your people-centric epics to move people operations from a strategic maturity phase to a transformational one?

- A) Adopt DevOps processes and acquire clouds skills for microservices, serverless computing, and containers**
- B) Clearly define Key Performance Indicators (KPIs) and objectives supporting cloud adoption**
- C) Have autonomy in teams and incorporate cloud skills from IT into business teams like data scientist and TensorFlow**
- D) Keep the company siloed and work on foundational skills like virtual machine provisioning**
-
-
-
-

Explanation

You will have autonomy in teams and incorporate cloud skills from IT into business teams like data scientist and TensorFlow to move to a transformational maturity phase. The transformational phase in people operations is characterized by removing role separations that existed previously in favor of more autonomous teams and cloud skills shared from IT to business teams. This ensures IT acts as an innovator for the business and provides key technology skills for employees that can use them.

You would not keep the company siloed and work on foundational skills like virtual machine provisioning to move to a transformational maturity phase. This is a feature of the tactical maturity phase of cloud adoption. In this phase, there is a focus on foundational cloud skills like budgeting and procurement. The tactical maturity phase also features a lack of measuring success.

You would not adopt DevOps processes and acquire clouds skill for microservices, serverless computing, and containers to move to a transformational maturity phase. This is a feature of the strategic phase of cloud adoption. The strategic maturity phase is characterized by an organization beginning to make changes for cloud adoption, which includes greater efficiency in IT teams and the involvement of people and processes in making use of the cloud for operations.

Clearly defining Key Performance Indicators (KPIs) and objectives supporting cloud adoption is not a feature of the transformational maturity phase but of the strategic phase of cloud adoption. The strategic phase covers change to the operations of the IT organization through the adoption of various DevOps practices.

The evolution of people-related epics through various Cloud Adoption Framework maturity phases occurs across various areas:

- Sponsorship – A strategy for cloud adoption is prioritized by leaders, and necessary resources are provided.
- Teamwork – Cloud technology adoption is optimized through trust-oriented and collaborative activities by teams.
- Communication – Communication is made transparent, and failures are shared with an intent to use them as a way to encourage further growth.
- Behaviors – Support is made available for employees who want to improve their teamwork and level of empathy for their customers.
- People Operations – Role definitions, measurement of performance, and employee recognition methods are used to empower task fulfillment by those who accept cloud adoption.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > Managing Change in the Cloud](#)

Question #41 of 60

Question ID: 1495908

You work as a cloud consultant for a manufacturing firm that wants to use cloud-based business intelligence (BI) tools to identify issues in its production lines. Which of these would you need to do to ensure that your company makes the best use of its BI tools?

- A) Opt for an on-premises BI tool for future scalability**
- B) Normalize and clean the data for BI**
- C) Encourage all users to use open source BI tools**
- D) Keep data for BI siloed in different systems**

Explanation

You would need to normalize and clean the data for business intelligence (BI). For data to be useful for BI, you need to ensure it is high quality, updated, and cleaned. BI is a process of using technology to collect and study organizational data for making strategic organizational decisions. This involves collecting the data into a storage repository like a data warehouse and using tools to perform data analysis on the stored data. Looker is a BI tool offered by Google Cloud. Some examples of BI applications include analysis of the shopping preferences of customers, costs of company operations, and sales data. BI is also useful for comparing a company's operations versus pre-established standards and baselines. There are three categories of BI tools:

- On-premises – These are used inside an on-premises data center with data warehouses also on-premises. They lack the scalability available in cloud-based BI tools.
- Open source – These are affordable tools and can help save on the costs of infrastructure when used as cloud-based variants. However, they need considerable expertise and coding skills to be used.
- Cloud-based – These are highly effective tools for working with data in high volumes and streaming formats. They are also economical as the technical expertise and infrastructural overhead required for their use is the responsibility of the cloud service provider (CSP).

You would not keep data for BI siloed in different systems as this does not allow users access to all of the data required for BI purposes. Siloed data is distributed across multiple systems and cannot be accessed by other systems due to outdated software or stringent user access permissions. This makes key data needed for BI become inaccessible. Using an Extract, Transform, and Load (ETL) tool can help consolidate the data from across multiple systems needed for BI applications.

You would not encourage all users to use open source BI tools. You need to use managed tools for BI as much as possible because it reduces the complexity of using the tools and ensures that a lack of programming expertise or technical know-how does not become a performance bottleneck in the BI process. Open source BI tools require technical knowledge and may not be suitable for all users inside an organization. You would instead opt for a managed cloud-based BI tool, such as Looker.

You would not opt for an on-premises BI tool for future scalability because on-premises systems are not as scalable as cloud-based solutions. For accommodating future growth and working with massive datasets, you need a cloud-based solution that allows you access to managed data warehouses and BI tools with various features built in for ease of use.

You need to know of some key ways an organization can utilize BI tools, including:

- Reporting of key data for creating strategy and initiatives for the business
- Mining data through massive datasets for identifying trends and patterns
- Providing visualization of data for easy comprehension of complex data
- Analyzing real-time and streaming data, which may be gathered from sources, such as Internet of Things (IoT) sensors and stock market data
- Using predictive modeling and analysis for forecasting future trends using historical data
- Measuring business performance by analyzing sales data and customer feedback

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > What is Business Intelligence?](#)

Question #42 of 60

Question ID: 1497405

You want to ensure maximum security of your sensitive data and also leverage cloud-native services like serverless computing. Which cloud deployment should you opt for?

- A) Private Cloud**
- B) Public Cloud**
- C) Hybrid Cloud**
- D) Multi-Cloud**

Explanation

You should opt for a hybrid cloud infrastructure model. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud always contains a private cloud. A hybrid cloud deployment allows a company to run its applications on-premises and use the public cloud for bursting when user demands go beyond the limits of the data center.

You would not opt for a private cloud infrastructure. In this deployment, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premises data center. This deployment option does not include systems that are a part of a public cloud service offering.

You would not opt for a public cloud infrastructure. In this deployment, the infrastructure includes applications that are fully functional in a cloud and have no on-premises portions. This deployment option may not be suitable for situations where sensitive data needs to be stored locally and not on a public medium.

You would not opt for a multi-cloud infrastructure. In this deployment, the data is still stored on a public system, which can cause security concerns for the company. A multi-cloud deployment involves using multiple public CSPs where each public cloud provides a different service for the company. This allows a company to use one public cloud for database services and another one for analytics. This deployment option lets a company use a preferred cloud provider for specific services. Remember that using multiple public CSPs indicates a multi-cloud infrastructure.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Question #43 of 60

Question ID: 1495903

A company is currently considering cloud adoption with its workloads operating individually without a collective strategy for future scalability. The company wants to keep its costs of operation as low as it can and move to the cloud with minimal possible interruptions to its operations. Based on this data, which cloud maturity phase of the Cloud Adoption Framework would you place the company at?

- A) Scale

B) Transformational

C) Tactical

D) Strategic
- ✖

✖

✔

✖

Explanation

Based on the assessment, the company is currently at the tactical phase of the Google Cloud Adoption Framework (CAF) which is characterized by a need to minimize current operational costs and make minimal changes to existing IT teams, technology, and processes. An organization in a tactical phase can seek to optimize costs by making better use of compute and storage resources that are currently underutilized. It can also seek to remove the delays and overhead associated with manual procurement and provisioning of resources.

The company is not at the strategic phase of the CAF. The strategic phase is characterized by efforts to modernize the architectures of a company's IT solution by using cloud-native services. This phase also includes some changes made to the existing IT teams, technology, and processes.

The company is not at the transformational phase of the CAF. The transformational phase is characterized by efforts to fully reorganize the IT organization of the company and enable more knowledge sharing to ensure individuals and teams can make more decisions with autonomy. This phase has a cloud-first approach by seeking to leverage the best cloud services and best practices as per current norms. The transformational phase helps meet a long-term objective for a business to convert its IT systems into an engine of innovation that enables business transformation.

Scale is not a phase of the CAF but is a theme for cloud adoption. There are four cloud adoption themes as per the CAF which include learn, lead, scale, and secure. The learn theme measures the learning systems in place for an organization to upskill its IT workforce. The lead theme measures the level of support that the IT workforce receives from the company's leadership for cloud migration. The scale theme measures the depth of usage of services that are native to the cloud for reducing overhead in operations as well as the level of automation you perform for policies and processes which were operated manually. The secure theme measures the level of protection in place for your environment from malicious access using a security system that has multiple layers and is identity-centric.

The three maturity phases component of the CAF gauge the readiness of a company for successful cloud adoption, which depends on its current business practices in each theme. These business practices can be in one of three possible phases:

- Tactical – This is where workloads are currently operating individually but not cohesively with a strategy for future growth. At this phase, a company is focused on keeping costs of individual systems as low as possible and moving onto a cloud system with the least possible disruption of operations. At this phase there is no provisioning done for future scalability.
- Strategic – This is when a wider strategy is used to govern individual workloads with an intent to scale to future needs. This is when a company has started accepting the changes necessary for cloud adoption and has involved its processes and people. At this phase, the company's IT department has efficiency and effectiveness which ensure greater value for a company through its cloud adoption for business.
- Transformational – This is where a company has its cloud operations running correctly and there is a focus on incorporating insights and data gathered from using the cloud. At this phase, a company undergoes transparent sharing of data, gathering and analysis of new data, use of machine learning (ML) for predictive and prescriptive analytical functions, and a transformation of both people and processes. This fosters further changes to technology. Additionally, the IT systems for a company go from being a cost center to a business partner and an engine for innovation. This phase also features blameless analysis at the C-level and development teams that are autonomous.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Question #44 of 60

Question ID: 1497147

You are a cloud services consultant for a digital publishing company that provides the news by covering the latest world events through web and mobile apps. You want to find a way for the news to be made available in several local languages across the world automatically. Which of these services can you use for this?

- A) Cloud Natural Language API

B) Cloud Translation

C) Cloud Vision

D) Cloud Spanner
- ✖

✔

✖

✖

Explanation

You would use Cloud Translation for this scenario. Cloud Translation or Translation Artificial Intelligence (AI) performs translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language or vice versa. By using Cloud Translation API, you can provide automatic text translation features for an app or Website.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers for understanding image content through machine learning (ML) models accessed through an API. It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images. Cloud Vision allows you to keep moderation for image content, use analysis of image sentiment for marketing strategies, and create metadata for an image catalog. It also allows you to perform image analysis for images uploaded on requests and can be integrated with Cloud Storage.

You would not use the Cloud Natural Language application programming interface (API) for this scenario. Cloud Natural Language API or Natural Language AI enables you to gain an understanding of natural language using API. This can help you identify the following aspects in a block of text:

- Key entities
- Positive or negative sentiments
- Language used
- Syntax including dependency trees and speech parts

The Cloud Natural Language API can be utilized by developers either through API calls or through reference to a document that may be stored on Cloud Storage.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner to provision a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Translation AI](#)

Question #45 of 60

Question ID: 1495222

The company you work for wants to find more secure and cost-effective methods of managing its systems. You are looking to select the appropriate cloud deployment for this. Which of these deployment configurations uses a hybrid cloud infrastructure?

- A) Using an on-premises data center for storing raw data along with a public cloud for data analytics**
- B)** Using one public cloud service provider for a web frontend and a different one for the backend
- C)** Storing all organizational data and applications in an on-premises data center
- D)** Using SaaS from one cloud provider and PaaS from another

Explanation

Using an on-premises data center for storing raw data along with a public cloud for data analytics is an example of a hybrid cloud infrastructure. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud also always contains a private cloud. A hybrid cloud deployment allows a company to run its applications on-premises and use the public cloud for bursting when user demands go beyond the limits of the data center.

Using one public cloud service provider (CSP) for a web frontend and a different one for the backend is not an example of a hybrid cloud deployment but of a multi-cloud deployment. A multi-cloud deployment involves using multiple public CSPs where each public cloud services a different function for the company. A company can then use one public cloud for database services and another one for analytics. This lets a company use its preferred cloud provider for specific services. Remember that using multiple public cloud service providers indicates a multi-cloud infrastructure.

Using Software as a Service (SaaS) from one cloud provider and Platform as a Service (PaaS) from another is not an example of a hybrid cloud infrastructure but of a multi-cloud infrastructure. This is because both SaaS and PaaS are public cloud offerings, and using several public clouds is a feature of multi-cloud deployments. The PaaS model allows consumers to install their own applications onto a cloud platform. The provider supplies the necessary operating system and physical hardware. This allows consumers to quickly deploy their own applications without the burden of first acquiring the necessary physical resources and operating system(s) to make it happen. The SaaS model covers the use of software applications by consumers who are running on a provider's cloud infrastructure. These applications can be accessed using thin client interfaces, including mobile devices and web browsers. An example of this is email accessible from a web browser, such as Gmail. The advantage of this model is that a consumer can utilize the application without the need to physically install the application or be concerned about the infrastructure needed to run it.

Storing all organizational data and applications in an on-premises data center is not an example of a hybrid cloud infrastructure but of a private cloud infrastructure. In this infrastructure, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premise data center.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Question #46 of 60

Question ID: 1497140

You need to create a system that can provide intuitive and high quality conversations for customers through voicebots and chatbots. What Google technology would you select for this need?

- A)** Media Translation API
- B)** Visual Inspection API
- C) Dialogflow**
- D)** Document AI

Explanation

You would use Dialogflow in this scenario. Dialogflow provides a suite for the development of conversational apps with voice and text capabilities, which include voicebots as well as chatbots. Dialogflow provides cross-platform connectivity to existing apps for the Internet of Things (IoT), iOS, or Android, and for telephony platforms, including Cisco, Genesys, and Avaya. It can also be used for digital platforms, including Slack, Facebook Messenger, and Google. The Dialogflow Customer Experience Edition (CX) enables the advanced development of chatbots and voicebots. It contains a visual platform for building bots, tools for modularizing bots, support for advanced interactive voice response (IVR), as well as tools for versioning and collaboration. It has been optimized and scaled for use at an enterprise-level.

Document Artificial Intelligence (AI) is not the appropriate choice for this scenario. You would use Document AI for the classification and extraction of structured data from existing documents. This can enable you to find key insights and automate organizational practices. You can use Human-in-the-Loop AI for workflow systems and a user interface to verify the information extracted from documents using Document AI.

Media Translation application programming interface (API) is not the appropriate choice for this scenario. You would use Media Translation API for automatic translation of audio between languages. This can also be done in real-time, if required. You can use Media Translation API for programmatic translation of audio inside of your applications. Media Translation API is based on the open source Google Remote Procedure Call (gPRC) framework.

You would not use Visual Inspection API for this scenario. You would use Visual Inspection API for training and deploying AI models for automatic detection, classification, and localization of any anomalies that are identified in images. This helps increase the quality of production in various industries and enables development of advanced analytics.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Dialogflow](#)

Question #47 of 60

Question ID: 1495277

To ensure the security of your cloud deployment, senior management has decided to implement the principle of least privilege for specifying user access permissions across the cloud. What will be the impact of using a resource access model based on the principle of least privilege for the security of the cloud?

- A) A user only has access to the cloud resources necessary for task completion

✓
- B) Only senior management can access resources on the cloud

✗
- C) Users need authorization to use on-premises applications

✗
- D) Cloud resource access needs to be approved by the cloud service provider

✗

Explanation

The impact of the least privilege principle in access management is that a user only has access to the cloud resources necessary for task completion. This ensures that users cannot access resources that are outside their scope of work, thus increasing privacy and security on the cloud. This access model ensures that users with malicious intent cannot cause damage to resources outside of their privileges.

The least privilege principle does not imply that only senior management can access resources on the cloud. An access management system that only allows senior management access to cloud resources would not allow for normal use of a cloud deployment and would not provide the company with the growth it seeks from using the cloud. By the principle of least privilege, all employees, regardless of their designation, can have access that is limited by their specific role within the organization.

The least privilege principle does not imply that users need authorization to use on-premises applications. The access in systems in place for an on-premises data center is not related to the Identity and Access Management (IAM) that are used on a cloud deployment for restricting access. The purpose of the IAM systems on a cloud is to keep a unified and secure system for authorizing user access to cloud resources.

The least privilege principle does not imply that cloud resource access needs to be approved by the cloud service provider (CSP). Specifying the access rights to cloud resources for users is not the responsibility of the CSP, like Google, but of the company that is using the cloud. As per the shared responsibility model, the CSP is responsible for the security of the cloud, and the customer is responsible for security in the cloud, which includes managing access rights for users.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Services.google.com > Google Cloud security foundations guide](#)

Question #48 of 60

Question ID: 1495926

Which Google Cloud solution can help your company modernize its infrastructure by removing downtime, reducing RTO, and preventing data corruption and loss?

- A) Cloud Run

✗
- B) Cloud Build

✗
- C) BigQuery

✗
- D) Cloud Backup and DR

✓

Explanation

You should use Cloud Backup and Disaster Recovery (DR) to meet these requirements. This service provides a way to backup workloads from both Google Cloud and on-premises systems for data protection that is consistent as well as centralized. It allows you to have a single system to protect file systems, database instances, and virtual machines (VMs) for both VMware and Compute Engine. Using application-consistent backups that are incremental-forever, Cloud Backup and DR ensures low values for both the recovery point objective (RPO) and recovery time objective (RTO) of a company, ensuring swift recovery of business operations using backups. This is enabled by the ability to instantly mount and use VMware engines and databases from Cloud Backup and DR backups. The incremental-forever system used by Cloud Backup and DR ensures that duplicate data does not need to be stored and results in reduced cost of storage and faster recovery times. This is made possible through changed-block tracking, which reduces storage requirements and speeds up recovery.

You would not use Cloud Run for this scenario. You would use Cloud Run when you need to perform a deployment of containerized applications, which are written in your preferred coding language. Cloud Run provides support for Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling.

You would not use Cloud Build for this scenario. Cloud Build is used to build, test, and deploy systems using a continuous integration and continuous delivery (CI/CD) platform, which is serverless. It allows a company to integrate its existing legacy systems and tools into a modern build system. This is made possible using third-party applications and customized steps for builds. Cloud Build also has vulnerability scanning features for CI/CD, which prevents the deployment of any images that are unsafe. Some key features of Cloud Build include:

- Fast build time using VMs with fast CPUs provisioned to cache source code and dependencies.
- Deployment automation using pipelines with integrated use of Firebase, Google Kubernetes Engine (GKE), Cloud Functions, and App Engine.
- Multicloud deployment using multilanguage support for builder images used and containerized tasks, which are portable across clouds.
- Strong privacy through Google Cloud security, which allows the use of repositories for private source code inside private networks.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use and has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Backup and DR Service](#)

Question #49 of 60

Question ID: 1495235

You want to perform refactoring of your existing on-premises systems so you can move workloads to the cloud while making key improvements. Which of these actions should you perform for this?

- A) Discard VMs and procure physical servers

✖
- B) Convert apps on VMs into containers on GKE

✔
- C) Move VMs from on-premises to Compute Engine

✖
- D) Extend VMware workloads using VMware Engine

✖

Explanation

You should convert apps on virtual machines (VMs) into containers on Google Kubernetes Engine (GKE). This is an example of refactoring, which involves moving and improving existing systems. The purpose of refactoring is to convert current workloads so that they can run on modern cloud systems without making any modifications to existing code. For this, you perform an automatic transformation of apps that are currently running on VMs into containers on GKE, convert databases into Cloud Spanner, and change data warehouses to BigQuery. This allows you to gain substantial improvements in workload performance by leveraging cloud-native systems.

You would not move VMs from on-premises to Compute Engine. This is not an example of refactoring but of replatforming, which involves lifting and optimizing existing systems by running them on the Cloud without code rewrites or changes. For this, you move all the workloads that utilize VMs from hypervisors that are on-premises or on other cloud systems onto Compute Engine on Google Cloud. This helps you gain cost optimization and maximization of performance using Google Cloud recommender and cost management tools.

You would not extend VMware workloads using VMware Engine. This is not an example of refactoring but of rehosting, which involves lifting and shifting existing systems onto a cloud with no changes to apps or the platform they run on. For this, you can run existing VMware workloads by utilizing VMware Engine, which effectively extends an on-premises system onto the Cloud. The purpose of this move is to ensure business continuity by use of existing systems and policies without any modifications and yet have a centralized way of managing your environment.

You would not discard VMs and procure physical servers. Procuring physical servers is not an example of infrastructure modernization. The purpose of infrastructure modernization is to move existing workloads from a data center-based computing environment, which uses physical servers to a virtualized cloud-based system for more elasticity, resiliency, availability, and scalability. You may consider procuring physical servers for an on-premises system under very specific circumstances, such as when certain data needs to be stored inside a company's facility for security and compliance requirements.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Data center migration into cloud](#)

Question #50 of 60

Question ID: 1497152

Your company has a large catalog with many stock keeping units (SKUs). The company wants to utilize efficient third-party tools for managing costs for projects programmatically in order to have greater clarity into cloud costs and their management. Which of these options will help meet this requirement?

- A) Billing the on-premises data center

✖
- B) Setting resource quotas

✖
- C) Billing the cloud service provider

✖
- D) Using billing APIs

✔

Explanation

Using billing application programming interfaces (APIs) will help meet this requirement. With billing APIs, you can use various tools created by other vendors to manage project bills programmatically and obtain data on all stock keeping units (SKUs) in their repository and related metadata. Other methods of ensuring clarity for financial governance include using intelligence recommendations for performing rightsizing of computing resources. This helps ensure that costs are minimized by ensuring that virtual machine (VM) usage is optimized. Also, you can use billing reports from the Google Cloud Platform (GCP) console to plan, track, and ensure optimal use of resources within your cloud deployment. This is done through views of trends for current costs and forecasts for future spending. You can also export this data to Data Studio and BigQuery to create customized dashboards that provide more details on cloud utilization and its associated costs.

You would not bill the on-premises data center in this scenario. This is not a way to provide programmatic management of project billing for a cloud deployment. For programmatic management of costs and billing, you need to utilize APIs, which can provide an interface between your systems and the services offered by third parties. The on-premises data center's operations and cost management are unrelated to the cloud deployment.

You would not set resource quotas in this scenario. Setting resource quotas does not provide a way for programmatically managing billing for projects. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budgets.

You will not bill the cloud service provider (CSP). You, as a customer, are billed by the CSP based on the resource consumption and service utilization of your cloud deployment. CSPs use the Pay-As-You-Go (PAYG) model of payment, where a cloud consumer only needs to pay for resources consumed and not for unused resources. For example, if a computing instance is in a terminated state, then it will not contribute to the total billable costs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Question #51 of 60

Question ID: 1495186

An e-commerce company is expanding its workforce and is working on performing data analysis to find ways to grow its business. You need to use a suitable service to improve decision making and productivity in the organization by providing key insights to a growing number of users. Which tool should you use for this requirement?

A) Cloud Trace 

B) Dataproc 

C) Dataflow 

D) **Looker** 

Explanation

You should use Looker. Looker is a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Dataflow for this scenario. Dataflow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

You would not use Cloud Trace in this scenario as it is not for BI, but a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Looker](#)

Question #52 of 60

Question ID: 1495931

A company uses custom made software to provide business services for consumers. The company is having trouble related to revealing certain aspects of its third-party applications to customers. How can it prevent the leakage of these vendor-related interfaces to its customers?

A) **Use API endpoints and move the code to the cloud** 

B) Block all exposed APIs to ensure confidentiality 

C) Allow customers to install vendor-specific apps 

D) Use a monolithic self-made application with no APIs 

Explanation

The company needs to use application programming interface (API) endpoints and move the code to the cloud. By utilizing endpoints that are vendor-agnostic and moving the entire application to Google Cloud Platform (GCP), you can ensure that any interfaces that are specific to vendors and third parties are not revealed to customers. You can move the application code to GCP using Migrate for Anthos.

The company should not allow customers to install vendor specific apps. You need to maintain a level of abstraction between customers and the vendors that create the applications you use to provide services to customers. You need to ensure that the customers do not have direct access to these applications and instead only reveal the necessary services to them using APIs.

The company should not use a monolithic self-made application with no APIs. APIs help move away from a monolithic architecture to application development and move to a microservices-based architecture. This provides greater stability to the system and ensures higher availability as individual components of the system can run independently, and there is no dependency between them. This ensures that if one component fails, then the other application components are not affected.

The company should not block all exposed APIs to ensure confidentiality. Consumers need access to APIs to perform various business tasks, so blocking access to the APIs would disrupt business operations. Instead, you need to provide a more controlled solution by first creating APIs that are vendor-agnostic. With APIs for various business services, multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Question #53 of 60

Question ID: 1495230

Your organization uses virtual machines for their main database processing application. Each time new capabilities need to be added to the application, the system needs to be brought offline, which causes significant downtime. What would you suggest as a more robust solution for this situation?

- A) Use high performance solid-state drives for storing application data

✖
- B) Use containers to compartmentalize the application

✔
- C) Upgrade the CPUs for the underlying infrastructure

✖
- D) Install updated hypervisors for running virtual machines

✖

Explanation

You should suggest using containers to compartmentalize the application in this scenario. By compartmentalizing an application using containers, you can then change parts of the application in isolation from the other parts. Containers are software packages that contain application code as well as dependencies, such as runtime environments and libraries, that are required for running an application. This allows containers to virtualize an OS and allows applications to be made portable across multiple systems. Google search, YouTube, and Gmail are all containerized applications. Containers allow you to:

- Separate responsibilities – This allows development teams to work on application logic and features and operations teams to focus on application deployment.
- Make workloads portable – This allows containers to be run from any OS like Windows, Mac, or Linux and on any system including virtual machines (VMs), physical servers, a developer's laptop, on-premises systems, or a public cloud.
- Isolate applications – Containers perform OS-level virtualization of resources like memory, CPU, storage media, and networking systems which isolates the OS from any other applications.

A key difference between containers and VMs is that containers perform virtualization at the level of the OS while VMs perform virtualization at the level of hardware.

You would not use high performance solid-state drives (SSDs) for storing application data. The storage medium for the system is not a contributing factor in performing application compartmentalization, which this scenario requires. You can use a variety of storage media for your system which can include hard disk drives (HDDs) or SSDs depending on your specific requirements.

You would not install updated hypervisors for running VMs. Hypervisors allow VMs to optimally execute several applications on VMs concurrently. However, they do not have provisions for compartmentalizing applications.

You would not upgrade the CPUs for the underlying infrastructure. Changing the CPUs can increase the processing capabilities of the underlying system but this will not alter the application architecture which needs to be compartmentalized using container technology.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > What are Containers?](#)

Question #54 of 60

Question ID: 1497154

Your company's cloud deployment is used by several teams and departments. You need to find an efficient way to ensure that the operating costs of the cloud deployment stays within the set budgetary limit specified by management. Which of these actions should you take to accomplish this?

- A) Use cloud resource labels and hierarchies

✖
- B) Set resource quotas and perform throttling

✔
- C) Manage inventory for on-premises systems

✖
- D) Use organizational policies for resource access

✖

Explanation

You should set resource quotas and perform throttling. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budget. Using Google Cloud tools, you can perform automated throttling and capping of resource consumption as well as broadcast budget notification alerts on company communication channels, such as Slack. The most important capabilities for cost management on the cloud include:

- Maintaining costs within budgetary limits
- Finding ways to save on IT costs
- Performing forecasting for IT expenditure
- Maintaining consistency in IT costs and keep deviations limited

You would not use organizational policies for resource access in this scenario. You would use these policies to gain more control over your cloud costs. The purpose of using organizational policies is to specify permissions for users to perform spending on Google Cloud based on their role in the organizational hierarchy. This way, you control the ability of specific users to view costs and perform administrative functions using Cloud Identity and Access Management (Cloud IAM) for cloud resources.

You would not use cloud resource labels and hierarchies in this scenario. You can achieve greater accountability in your cloud deployment for financial governance by using cloud resource labels and hierarchies. Using resource hierarchies simplifies the process of allocating costs to specific departments to ensure accountability. This also provides insights into the return on investment (ROI) of the investment you have made on the cloud.

Managing inventory for on-premises systems is not a way of implementing budgetary control on a cloud deployment. An on-premises system and its inventory are independent of cloud resources. The financial management of on-premises systems is not related to the financial governance systems that need to be implemented to manage costs on a cloud deployment.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Question #55 of 60

Question ID: 1495255

Your company offers data analytics services using resources on its on-premises systems as well as its cloud deployment. It is now looking for a way to host its APIs both on-premises and on Google Cloud. Which Google Cloud service should you recommend for this?

- A) Cloud Backup and DR

✖
- B) Dataproc

✖
- C) Cloud Trace

✖
- D) Apigee hybrid

✔

Explanation

You should recommend Apigee hybrid for this scenario. With Apigee hybrid, you can place gateways near the source of application programming interface (API) traffic to use existing governance and security infrastructure while also benefiting from cloud-native services for monitoring and analytics. This is made possible by managing and controlling the runtime environment using Apigee hybrid. Apigee is used to build, secure, scale, and analyze APIs at any location with support for the complete API lifecycle. It provides strong systems of API control and visibility. With Apigee, you can increase the value of your digital assets by monetizing your API-based products.

You would not use Cloud Backup and Disaster Recovery (DR) in this scenario. Cloud Backup and DR is a service that provides a way to backup workloads from both Google Cloud and on-premises systems for data protection that is consistent as well as centralized. It allows you to have a single system to protect file systems, database instances, and virtual machines (VMs) for both VMware and Compute Engine. Using application-consistent backups that are incremental-forever, Cloud Backup and DR ensures low values for both the recovery point objective (RPO) and recovery time objective (RTO) of a company, ensuring swift recovery of business operations using backups.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Cloud Trace in this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Question #56 of 60

Question ID: 1495197

A company that develops database applications is looking for a Google Cloud service that will enable its applications to get better performance by incorporating the availability and scalability of Redis services without the associated complexities of managing the Redis deployments. Which service should you use in this scenario?

- A) Memorystore

✔
- B) Datastore

✖
- C) Firestore

✖
- D) Cloud Bigtable

✖

Explanation

You should use Memorystore in this scenario. Memorystore for Redis is a scalable, fully managed service for Redis on Google Cloud that provides 99.9% availability using up to five read replicas that are distributed across multiple availability zones (AZs). Memorystore provides automation for complex Redis and Memcached tasks, which include ensuring high availability, creating failovers, and performing patching. It is compatible with the protocols of both Memcached and Redis. Memorystore also has provision for private IP addresses and virtual private cloud networks (VPCs) for data security and has integration with Identity and Access Management (IAM) for access control. Memcached provides in-memory caching for temporary data storage. Redis is a key-value NoSQL database using an in-memory data system for high performance. Redis supports sharding and uses persistent data, while Memcached does not.

Firestore is not the appropriate choice for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the Internet, mobile, and the Internet of Things (IoT) that can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

Dataprep is not the appropriate choice for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for the purpose of machine learning (ML) and analysis. It is an intelligent data service that is serverless, so the user does not need to deploy or manage the infrastructure. Dataprep suggests the data transformations that are ideal each time a User Interface (UI) input is detected, so the user does not need to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. It helps the user avoid having to assess the quality of the data and proceed directly to exploratory data analysis.

Cloud Bigtable is not the appropriate choice for this scenario. Cloud Bigtable is a NoSQL database service that is fully managed and enables you to work with massive workloads for analytics and operations with availability at 99.999%. It is ideal for uses that include applications for IoT, finance, banking, personalized services, and advertising technology. Cloud Bigtable has a storage engine that allows for improved predictions in applications utilizing ML. It also has integration with tools for big data including Hadoop, Dataproc, and Dataflow.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Memorystore](#)

Question #57 of 60

Question ID: 1495943

A digital imaging company using Google Cloud has asked you to calculate downtime for its image processing service. The service has an availability SLO of 99.5% for a month. What is the error budget of the service?

A) 99%

B) 0%

C) 100%

D) 0.5%

✖

✖

✖

✔

Explanation

The error budget in this scenario is 0.5%. This is equivalent to about three and a half hours of downtime. The error budget is the measure of the system's possible downtime, which is the difference between the service level objective (SLO) and the total availability figure of 100%:

Error budget = 100% – SLO = 100% – 99.5% = 0.5%

When an SLO is agreed upon, its corresponding service downtime or error budget is also agreed upon. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

99% is not the error budget for this scenario. The value of 99% can be acceptable as the service level agreement (SLA) value. You need to keep an SLO value that is higher than the SLA, which helps you get early warnings when there is a danger of not meeting the SLA. SLAs are visible to customers that use a service while SLOs and SLIs are internal to the development and operations team. If you do not meet an SLO, then there is a possibility of not meeting the SLA which can cause issues for the company.

100% is not the error budget for this scenario. The value of 100% can be the desired value for reliability for a system, but it is not achievable in a real-world scenario. Costs and reliability need to be balanced in all real-world applications, and a system always depends on other systems, which reduces the level of reliability because the failure of one system can have an effect on others.

0% is not the error budget for this scenario. There is always a measurable value for an error budget because no system can have 100% uptime, and the error budget specifies the possible downtime that the system can face. When new features are added to a service, new bugs may be introduced, which can eat into an existing error budget. There can then come a stage where the error budget for a given period is exhausted, and management will need to halt development of new features until the error budget is restored. For example, when a company that deploys code changes twice a day consumes half its error budget, it needs to reduce code deployments to once a day. When three-fourths of the error budget is consumed, the deployments can be reduced to once a month. When the error budget hits zero for a month, deployments will need to be halted.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Question #58 of 60

Question ID: 1497406

Your operations and development teams need to create a plan for ensuring service availability while adding some key enhancements. For this purpose, which of these can you consider valid service level indicators (SLIs) to meet your service level objectives (SLOs)? (Choose two.)

A) Number of API calls generated for the service

B) CPU utilization for the service

C) Number of successful login attempts

D) Percentage of correct user responses

E) Amount of memory consumed by the service

✖

✖

✔

✔

✖

Explanation

The percentage of correct user responses and the number of successful login attempts are both valid service level indicators (SLIs) that you can use to meet your service level objectives (SLOs). SLIs measure the quantity of a specific aspect of the service that is relevant to the user. SLOs are defined based on user experience and can address user-centric needs, such as latency. SLIs help gauge if SLOs are met. A combination of SLOs and SLIs work towards meeting service level agreements (SLAs), which specify the exact agreement made with the users of the service.

CPU utilization for the service, the amount of memory consumed by the service, and the number of application programming interface (API) calls generated for the service are all not the best options for SLIs as these are internal to the system and not user-centric. By selecting SLIs that capture performance aspects that are relevant to the user, you can better meet SLA needs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Question #59 of 60

Question ID: 1495254

You work for a digital publishing company that owns millions of digital assets, including images, videos, and music. You want to find a way to monetize these digital assets to grow your business value. Which of these Google Cloud services would help you achieve this goal?

- A) Payment Gateway**
- B) Apigee**
- C) Dataflow**
- D) GKE**

Explanation

You should use Apigee to help you achieve this goal. With Apigee, you can increase the value of your digital assets by monetizing your application programming interfaces (API)-based products. Apigee is used for building, securing, scaling, and analyzing APIs at any location with support for the complete API lifecycle. It provides strong systems of API control and visibility. Apigee lets you monitor APIs using artificial intelligence (AI) powered systems. This ensures your APIs are delivering expected performance and maintaining high availability for customers and developers. You can also identify any API issues and diagnose them easily using the monitoring systems for Apigee. Additionally, Apigee allows you to use AI with your APIs historical metadata and perform behavior prediction as well as anomaly detection in real-time. Apigee also helps maintain regulatory compliance.

You would not use Payment Gateway for this scenario. With Payment Gateway, you can perform processing of payments that are API-based. It can scale up to cater to unpredictable surges in usage as well as be extended to third-party insights and anomaly finding tools through APIs. It also allows for payment processing from across multiple regions of Google Cloud's global infrastructure. Payment Gateway also has built-in tools that allow you to monitor, debug, and trace API-based purchases with high availability and less overhead. You can also export data of transactions into a project on Google Cloud using an export API.

You would not use Google Kubernetes Engine (GKE) for this scenario. GKE is not used for digital asset monetization, but it allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes, enabling you to specify CPU, RAM, storage, GPUs, and network configurations along with an Autopilot option for situations where there is not enough operational and developer support available.

You would not use Dataflow for this scenario. Dataflow provides an economical, swift, and serverless system for stream and batch processing work. It is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Question #60 of 60

Question ID: 1497157

Your company's accounting department has raised an issue regarding the high operating costs of the cloud deployment. You are asked to find ways to optimize the costs of running the cloud. What should you do to reduce the Total Cost of Ownership (TCO) using Google Cloud?

- A) Utilize historical usage data for monitoring**
- B) Delete idle VM instances**
- C) Use larger VM sizes than needed**
- D) Store data on the most available storage classes**

Explanation

You should reduce the Total Cost of Ownership (TCO) through cost optimization tactics, such as deleting idle virtual machine (VM) instances. Cost optimization involves using managed services, keeping a consumption model, and analyzing expenditure and attributing it correctly. Cloud monitoring dashboards display key system metrics graphically in real-time, offering an updated view of the cloud system. This helps you get a clear analysis of how your systems infrastructure, applications, and services are doing. This way, you can use metrics and logs to track and resolve issues swiftly, thus improving your TCO and reducing Mean Time to Recovery (MTTR). You can also create alarms based on metrics. An example is ensuring that server sizes are selected to match actual computing needs in order to ensure greater cost-effectiveness.

You would not reduce the TCO by using larger VM sizes than needed. Using larger server sizes than required will cause a waste of resources and thus a waste of capital. Instead, you can opt for rightsizing VM instances based on your specific requirements. For this, you can create a custom machine type that allows you to specify the exact CPU and RAM configuration for your specific processing needs.

You would not reduce the TCO by using historical usage data for monitoring. Historical data cannot give you an updated view of a cloud system. Monitoring involves observing important system metrics in real-time using dashboards. This allows you to find ways to optimize costs through more efficient use of cloud resources. An example is monitoring VM instances to identify resources that may be idle.

You would not reduce the TCO by storing data on the most available storage classes. Using the most available storage class is suitable for high availability applications. However, it is also the most expensive, and not all types of data require high availability. By using more suitable storage classes for specific types of data, you can optimize storage costs and save on TCO. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > Best practices for optimizing your cloud costs](#)